Astroparticle Physics

Mapping the Photon Detection Efficiency of VUV Sensitive SiPMs from the Ultra-Violet to the Near Infra-Red

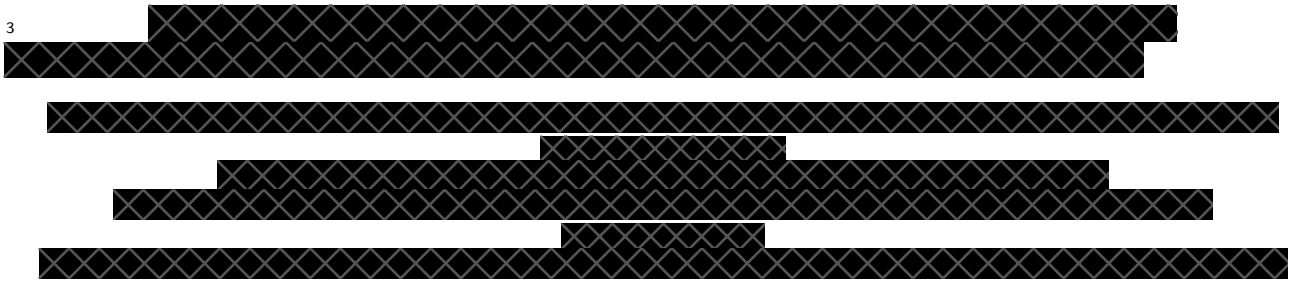
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Abstract

We present a flexible analytic model describing the Photo-Detection Efficiency (PDE) for P-on-N silicon photomultipliers. This model can be used for device characterization, PDE extrapolation from limited data or design optimization. The model uses six device specific parameters to describe PDE for a large range of wavelengths, angles of incidence, voltages and temperatures. Refractive index data taken from literature are inputs to the model, which also allows the prediction of PDE for devices operated in dense media. We present the measurement of the absolute PDE from 350-830 nm at 163 K for two UV sensitive devices: Hamamatsu VUV4 and the Fondazione Bruno Kessler VUV-HD3. Additional measurements of relative PDE versus angle are performed. We successfully fit the model to the data and show its predictive power, extrapolating the PDE to new wavelengths and to operation in liquid argon and xenon. Lastly we investigate optimizing efficiency for specific applications in astroparticle physics and quantum computing.

Keywords: SiPM, silicon photomultiplier, photodetection efficiency, PDE, liquid nobles, argon, xenon

1. Introduction

This work develops a flexible and comprehensive model for the photon detection efficiency (PDE) of silicon photomultipliers (SiPMs), which can be applied across their full range of detectable wavelengths and varied operating conditions. SiPMs have superseded the photomultiplier tube in many physics applications due to their lower operating voltage, improved radiopurity, compact size and physical durability. Devices consist of a pixelated array of Single Photon Avalanche Diodes (SPADs), operated in Geiger mode, which produce a characteristic charge pulse in response to a detected photon. SiPMs will be used to detect scintillation light from liquid Argon (LAr) and liquid Xenon (LXe) in noble liquid detectors such as nEXO, SBC, Darkside-20K, DUNE, and DARWIN/XLZD [1, 2, 3, 4, 5]. The model presented describes SiPM PDE as a function



of wavelength, incident angle, optical medium, temperature and overvoltage using optical data as fixed inputs. The model takes device specific input parameters describing the semiconductor junction structure, avalanche probability and optical characteristics of surface material.

A detailed understanding of wavelength-dependent PDE is essential in accurately characterizing SiPM response for future experiments. Complete measurement of PDE across the 5-variable parameter space is experimentally challenging; a descriptive model allowing confident extrapolation across phase space will reduce uncertainty in detector simulations. A full PDE model also enables accurate assessment of the experimental impact of external crosstalk (ExT), a nuisance parameter in SiPM based detectors. ExT occurs when secondary photons, which are produced during a SPAD's avalanche, escape the original SiPM and trigger other SiPMs [6]. External crosstalk is expected to be a contributing factor to the overall performance of noble liquid TPC detectors as it deteriorates energy resolution and can mimic low occupancy events. LAr and LXe emit scintillation light about the VUV wavelengths of 175 nm [7] and 127 nm [8], respectively. This necessitates the use of newly-developed VUV-sensitive SiPMs [9], which recent work [10] has shown emit ExT photons in the wavelength range of approximately 550-1050 nm. As such, a key motivation for producing this PDE model is to permit calculation of the probability of external crosstalk for SiPMs submerged in a liquid scintillator, derived from measurements taken in vacuum over a limited wavelength range.

The model is also intended to be a tool for designing SiPMs and single-pixel SPADs with optimized PDE. Higher detection efficiency leads to liquid noble detectors with improved energy resolution, lower event energy thresholds, and better sensitivity to new physics. In this work we propose device designs to improve detection efficiency to LAr scintillation, which is limited in current-generation devices [3]. Additionally, applications in quantum computing and quantum key distribution (QKD) require single-photon detectors with high PDE and timing resolution [11, 12, 13]. We discuss how devices may be optimized for efficiency at fixed wavelengths and provide parameters expected to yield PDE $> 80\%$ for specific applications.

The devices studied in this work are two VUV-sensitive SiPMs, the Hamamatsu (HPK) VUV4 and the Fondazione Bruno Kessler (FBK) VUV-HD3, which are used in the SBC experiment [1, 14] and are candidates for nEXO [15]. The PDE of the devices has been measured at wavelengths of 350-830 nm, and our model is used to extrapolate the PDE from 120 nm to 1050 nm. PDE as a function of angle has also been measured and is studied in the context of detailed optical modeling.

2. Experimental setup and methodology

Measurements were taken using the Vacuum Efficiency Reflectivity and Absorption (VERA) setup at TRIUMF, which is described in detail in [16]. A deuterium continuum lamp is used in combination with a vacuum monochromator to provide illumination. A NIST calibrated photodiode (Thorlabs FDS10X10) was used to measure the absolute photon flux of the lamp in the range 350-830 nm. Cooling was provided by liquid nitrogen supplied to a cold finger, with the temperature stable within ± 0.1 K at the nominal measurement temperature of 163 K (LXe temperature). Current measurements were performed using a Keysight B2985A picoammeter. For determining pulse rates at the waveform level, the output of each SiPM was converted to a voltage pulse using a custom-made transimpedance amplifier and digitized using a CAEN DT5730 digitizer. The angle of incident light upon the sample holder was varied by a stepper motor within a range of motion of -15° to 60° wrt to normal.

The spread of wavelengths emitted from the monochromator (MC) is incorporated in the analysis by triangular convolution of the PDE function with respect to wavelength, as expected for the transfer function of a simple diffraction grating based monochromator. This treatment assumes a flat spectrum is input to the MC. For wavelengths close to peaks in the output spectrum of the lamp this assumption is not valid and contributes to systematic uncertainties for certain wavelengths. The imact of this is discussed further in subsection 4.6. Data was taken with wavelength the mechanical slit controlling the FWHM values between 0.5-2 mm, using a previously measured scaling of 4.5 nm/mm. This scaling was treated as a nuisance parameter in the analysis of the oscillating FBK $PDE(\theta)$ data (subsection 4.6) and varied to minimize the global chi-squared. The resulting monochromator scaling increasing by 31% to a value of 5.9 nm/mm. We believe this ellipsometric measurement improves the accuracy of the previously calibrated value and is applied in all analyses, the data's FWHM values are between 3-12 nm.

2.1. Absolute PDE

The absolute PDE was measured for a range of wavelengths, angles of incidence and overvoltages. Using the adjustable pinhole of the optical collimator, the beam spot was kept sufficiently small to be entirely contained within the active area of the device, except where otherwise noted in angular PDE measurements. For each SiPM the absolute responsivity R_{SiPM} was calculated by measuring the light induced photocurrent on the SiPM and calibrated diode, as shown in equation 1. Dark current was subtracted for measurements performed at warmer temperatures.

$$R_{SiPM}(\lambda) = \frac{I_{SiPM}(\lambda) - I_{D,SiPM}}{I_{Diode}(\lambda) - I_{D,Diode}} \cdot R_{diode}(\lambda) \tag{1}$$

Linear responsivity is then converted to photon detection efficiency using equation 2:

$$PDE(\lambda) = \frac{R_{SiPM}(\lambda) \cdot E_{photon}(\lambda)}{Q_a} \tag{2}$$

Where E_{photon} is the photon energy at a given wavelength and Q_a is the gross average charge per single photon detection events. Q_a is defined as the sum of the charge gain of a single photoelectron (SPE) pulse and the excess charge arising from correlated noise effects such as afterpulsing and internal cross-talk. Q_a is calculated by taking waveform-level data for a given SiPM and determining the rate of 'primary' pulses using the method described in [17]. A primary pulse is defined as a pulse generated by the detection of a photon or a dark pulse. A DC current measurement of the SiPM is then taken under the same operating conditions, and the current is divided by the rate of primary pulses to yield Q_a. This measurement is repeated for various voltages as $Q_a(V)$ increases non-linearly with V due to the gain increase combined with correlated noise. Although Q_a is not expected to vary with wavelength, measurements were taken in both the dark and under illumination at several wavelengths in order to corroborate the measured values, with uncertainty on Q_a propagated through the PDE.

2.2. Angular dependence of PDE

The relative change in PDE with respect to angle was measured in order to probe the optical properties of the device surfaces. This measurement is simpler than the absolute measurement as it is independent of the excess charge factor and thus can be performed using IV methods. It is also not limited to the wavelength range of the calibrated photodiode. These measurements were

taken by measuring DC current at fixed wavelengths from 190-830 nm and normalizing by the current at normal incidence, after dark current subtraction for warm datasets. An optical pinhole of either 2 mm or 0.5 mm was used, with a larger pinhole giving higher photocurrent but a more limited range of incidence angles before light is no longer contained or blocked by the edge of the device package. The maximum achievable angle for the VUV4 devices within VERA is between $30-50^{\circ}$ and slightly higher for the FBK, depending on the pinhole size and SiPM positioning.

2.3. Measuring electron avalanche triggering probability

The probability that an electron passing through the high-field region of the SiPM will trigger an avalanche, P_e, can be independently measured using the method described in [16]. In summary, UV light with sufficiently short absorption length is used to ensure that photogeneration of carriers is localized to the p-doped region of the device and only electron induced avalanches occur. The primary pulse rate under illumination (with the dark pulse rate subtracted) is measured at various overvoltages, and normalized by the rate at high overvoltage where P_e is assumed to saturate to unity. Normalizing by the saturated rate yields $P_e(V)$ as all other variables impacting the PDE are independent of V. The wavelength condition for complete absorption can be verified experimentally by measuring the saturation of pulse rate with overvoltage [18], as $P_h(V)$ does not saturate under normal SiPM operating conditions. In this work 380 nm light was used to measure $P_e(V)$ for the HPK device at various temperatures from 163-253 K, which showed negligible temperature dependence.

3. Modeling the photon detection efficiency

The experimental variables influencing PDE are photon wavelength (λ), incidence angle (θ), operating over-voltage (V) and to a lesser extent the temperature (T). Operation in dense media also modifies the PDE via optical coupling. We present a model for PDE using device specific parameters describing a simple PN junction[1] covered with a silicon-dioxide thin-film surface layer, and avalanche triggering probabilities of minority carriers. The model presented pertains to p-on-n devices although it can be easily reversed for n-on-p devices; a simplified device structure is shown in figure 1. Listed below are the 6 model parameters and some defined mathematical quantities..

- t_{ox} - average thickness of the surface silicon dioxide (SiO_2) passivation layer.
- dp^* - effective depth below which electrons are collected, reaching the high field region (about X_{PN})
- X_{PN} - the center of the p-n junction and location of high field region, boundary where avalanches are either e- or h+ driven
- dw^* - effective depth above which holes are collected, reaching the high field region (about X_{PN})
- $P_e(V), P_h(V)$ - probability a collected electron, hole triggers an avalanche. Empirically parameterized wrt voltage as $P_e(V; A_e, V_e) = A_e(1 - e^{-\frac{V}{V_e}})$ and $P_h(V; A_h, V_h)$.
- FF - fraction of photo-sensitive surface area
- **J** - the set of device parameters listed above
- W_p, W_n - probability of photo-absorption within effective p, n regions
- T_{Si} - total probability of transmission into bulk silicon

[1] Naming conventions follow from [18]. Junction locations are wrt the silicon surface.

• $iPDE$ - the internal detection efficiency, efficiency for a photon transmitted into bulk silicon

The PDE equation is logically factored into the separate terms: FF (which may depend on AOI, subsection 3.4), transmission into silicon T_{Si}, and internal efficiency $iPDE$. The internal efficiency is given by the photoabsorption probability in the p (n) region W_p (W_n), multiplied by the avalanche triggering probability of the corresponding minority carrier P_e (P_h). The collection depths dp^* and dw^* include some diffusive contribution about the edges, thus are effective parameters correlated with the boundaries of the junction's depletion region. The PDE equation is given below, with the first line omitting arguments for clarity, while the second and third explicitly show the dependence on variables and parameters.

$$PDE = FF \cdot T \cdot (P_e \cdot W_p + P_h \cdot W_n) \tag{3}$$

$$PDE(\lambda, \theta, V; \mathbf{J}) = FF \cdot T_{Si}(\theta, \lambda; t_{ox}) \cdot iPDE(V, \lambda; \mathbf{J}) \tag{4}$$

$$iPDE(V, \lambda; \mathbf{J}) = P_e(V) W_p(\lambda; dp^*, X_{PN}) + P_h(V) W_n(\lambda; X_{PN}, dw^*) \tag{5}$$

To an excellent approximation the absorption functions W_p, W_n are independent of incident photon angle θ. In majority of cases the fill factor FF is fixed, while for precision modeling of certain devices FF may be a function of incident angle.

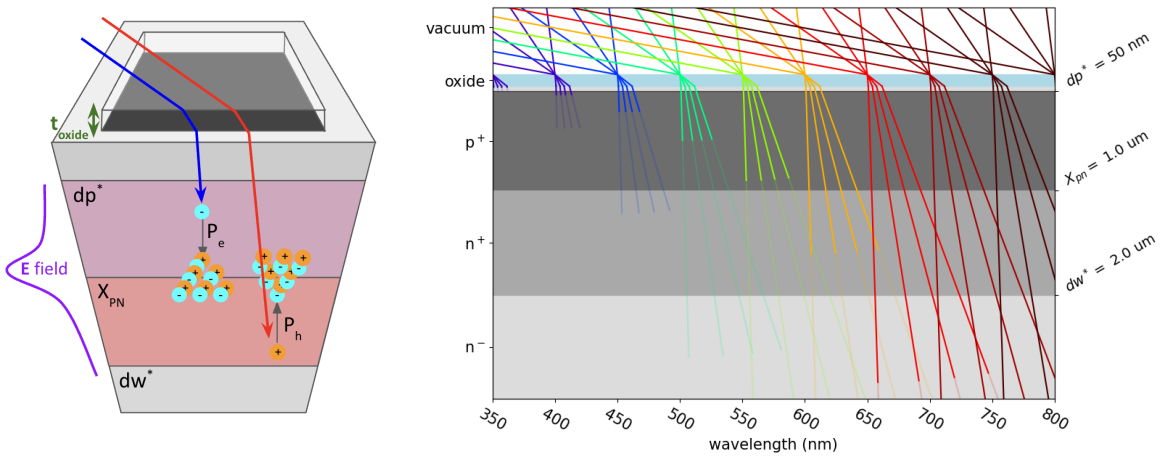


Figure 1: (Left) Diagram of the modeled SiPM structure, with illustration of the respective minority carriers drifting into the high field region initiating avalanches. (Right) Ray diagram of photon paths in SiPM (for real part of internal angle θ_3). Thick lines in silicon correspond to that wavelength's attenuation length $1/\mu$, with fainter lines drawn at length $4/\mu$. n^- denotes the insensitive bulk silicon. The refracted angles are drawn accurately, highlighting the similarity in path length within silicon for various angles of incidence.

3.1. Input optical data

The complex refractive indices of silicon, SiO_2 and detection media are fixed inputs to the model, allowing the calculation of the terms T_{Si}, W_p and W_n in Equation 4. To cover the wavelength range of interest, the n and k data for pure silicon were combined together from [19] for the VUV (110-200 nm), [20] (200-750 nm) for UV-Vis and [21] into the IR. This is referred to as the 'default' silicon dataset. The SiO_2 data was taken from [19] from the VUV-800 nm and [22] into the IR[2]. In [23]

[2]Datasets were merged together and smoothed in a 50 nm region of overlap when there was small disagreement between values.

Franta *et al.* presents a complex theoretical model applied to a wide range of optical measurements, yielding n and k values as a function of silicon purity and temperature. Curves for ultrapure 'float zone' silicon at 150 K are used[3]. This is referred to hereafter as the 'Franta' dataset. For the default dataset some ad-hoc selection was required; for SiO_2 we chose the dataset from [19] with $k = 0$ down to ~160 nm, consistent with measured PDE and expectations of transparency to LXe scintillation light. For Si, we consider the visible-range dataset from [20] to be more reliable than an alternative dataset from [21], which exhibited a shape inconsistent with the higher-precision Franta dataset and yielded a very poor quality of fit.

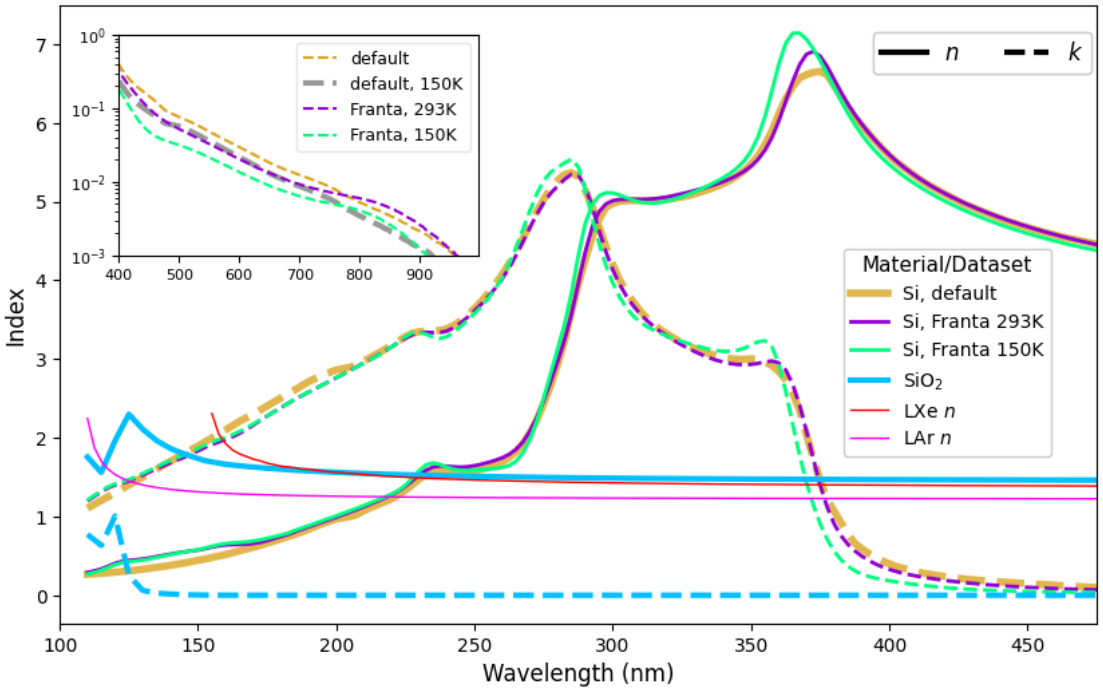


Figure 2: The input optical data used in this work. (Main figure) solid and dashed lines represent n and k, respectively. All silicon n values asymptote to ~3.55 at 1000 nm. The inset figure shows Vis-IR k curves on a log scale; the differences between these curves drive the difference in fit results between optical datasets. The thicker gray 'default, 150 K' k curve under the purple line is the default data scaled by the photoabsorption model. Data references are given in the text. For the liquid nobles n values are drawn where $k = 0$, and below the range of IR resonances reported in [24].

The input optical data is plotted in figure 2, including LXe data from [25], [26] and LAr data from [24]. Notably, in the NIR Franta has much lower k than the default dataset. Selecting data for pure silicon makes the simplifying assumption that modifications to $\bar{n}$ due to the p and n dopants are negligible, avoiding the complexity of estimating doping profiles. This assumption is revisited in the discussion and is emphasized in section 4, as usage of the Franta dataset significantly modifies fit parameters. A photoabsorptive temperature correction, described in subsection 3.5, does not reduce the room temperature k values of the default dataset as low as those of Franta.

[3] via https://refractiveindex.info/

3.2. Transmittance and reflectivity

The SiPM is modeled as pure bulk silicon (Si) with a thin silicon-dioxide (SiO_2) surface layer, following from the success of this approach in fitting SiPM reflectivity [27]. The vacuum-SiO_2-Si layers are labeled with subscript $1, 2, 3$ respectively. The complex refraction angles in the oxide θ_2 and silicon θ_3 are calculated using Snell's law, illustrated in figure 1 using $\Re(\theta_3)$. Equation 6 is the wave-optics transmission for a two layer system with a thin central film of thickness t_{ox} [28], giving the transmission from region 1 to 3 for S($\perp$) or P($\parallel$) polarized photons, and $r_{i,j}$, $t_{i,j}$ the Fresnel coefficients from medium i to j. The phase change due to the changes in optical path length, arising from internal reflections within the thin film, is given by δ which drives the oscillatory behavior of R, T and describes attenuation within the SiO_2 layer.

$$T_{13,\parallel or \perp}(\lambda, \theta; t_{ox}) = \frac{\Re(n_3 \cos\theta_3)}{\Re(n_1 \cos\theta_1)} \Big| \frac{t_{12} t_{23} e^{i\delta}}{1 + r_{12} r_{23} e^{i2\delta}} \Big|^2_{\parallel or \perp} \tag{6}$$

$$\delta = 2\pi \frac{n_2 t_{ox}}{\lambda} \cos\theta_2 \tag{7}$$

$$T_{Si} = 1/2 (T_{13,\parallel} + T_{13,\perp}) \tag{8}$$

For our application the two polarizations are averaged for the transmission into silicon. A similar expression for reflectivity is used in the analysis, taken from [27] and also averaged to give R_{Si}.

$$R_{13,\parallel or \perp}(\lambda, \theta; t_{ox}) = |\frac{r_{12} + r_{23} e^{i\delta}}{1 + r_{12} r_{23} e^{i\delta}}|^2_{\parallel or \perp} \tag{9}$$

An important distinction must be made for P-polarized light at oblique incidence on lossy media such as silicon. In this scenario, the Fresnel field equations predict $T + R < 1.0$, however using the Poynting vector to calculate the transmission retains $R + T = 1.0$ with the direction of energy flow bent further towards the surface normal than $\Re(\theta_3)$. The 'missing' Fresnel fraction is typically attributed to absorption within a skin layer, in the case of SiPMs this may contribute to the PDE. To include this missing fraction, for oblique incidence above 160 nm we use $T_{Si} = 1 - R_{Si}$. Equation 9 is valid for all phase space, and Equation 6 valid for normal incidence. For oblique angles and wavelengths below 160 nm, where absorption occurs in the quartz film, disentangling the missing fraction from absorption in SiO_2 is less straightforward; using Equation 6 with $\Re(\theta_3)$ is a good approximation but not exact.

The Hamamatsu VUV4 device has a thick quartz window above its surface. Using Fresnel's equations the silicon dioxide data is used to calculate the window's transmission and reflectivity T_W and R_W. Applying ray optics these coefficients are combined with T_{Si} and R_{Si} by the geometric series, including multiple reflections between the window and SiPM giving the total transmission for the system:

$$T_{total} = \frac{T_W T_{Si}}{1 - R_{Si} R_W} \tag{10}$$

3.3. Internal PDE

Once a photon enters the silicon, it can be absorbed in the 'dead region'[4] from $0 < x < dp^*$, in the p-doped region $dp^* < x < X_{PN}$, the n-doped region $X_{PN} < x < dw^*$, or in the insensitive

[4] (so-called because it is assumed photo-generated carriers produced close to the device surface will be lost due to recombination

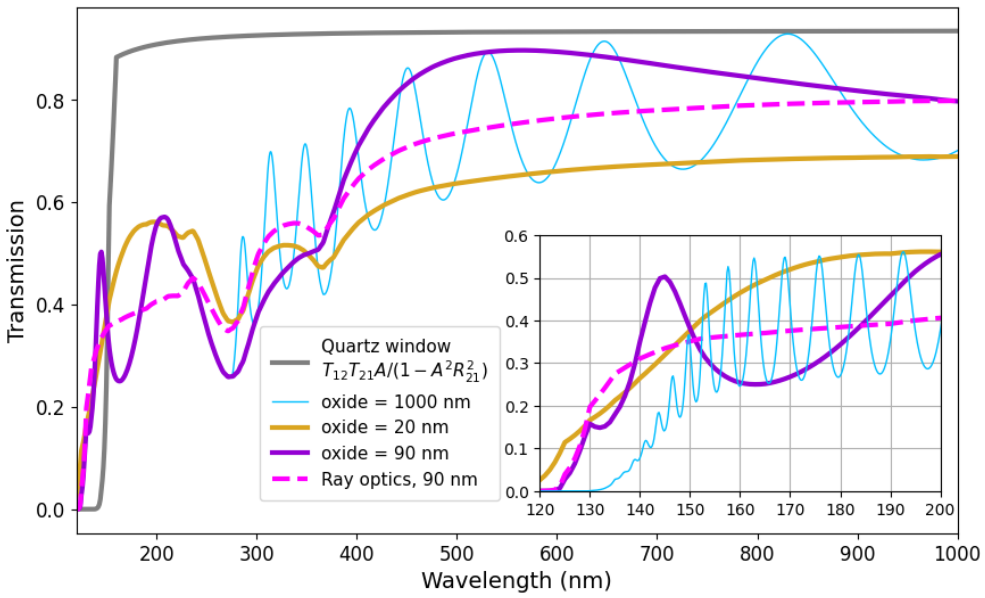


Figure 3: Vacuum transmission for the vacuum-quartz-silicon thin film interface used in the PDE model. The dashed pink line is the transmission for ray optics; the thin-film transmission (using Equation 6) oscillates about this mean value. The 20 nm (gold) transmission is interference-enhanced below 300 nm while the thicker 90 nm curve (purple) oscillates in the VUV and is constructive above 380 nm. The 1μm oxide oscillates quickly about the median (truncated in VUV for clarity). Inset graph shows the region of interest for liquid noble applications.

bulk. The silicon absorption length μ is calculated from the imaginary part of the refractive index, the extinction coefficient k, using the common relationship

$$\mu(\lambda; k) = 4\pi\frac{k}{\lambda} \tag{11}$$

The photon path lengths within each region are calculated using the internal angle θ_3. The photon absorption weights for each region of the diode, W_p and W_n, are the integral over the relevant region scaled by the attenuation prior to reaching that region, see Equation 12 and example curves in Figure 4. It is important to note that due to the strong refraction or 'lensing' into the silicon, the W values have a very weak dependence on θ_1 over the majority of phase space where transmission remains significant; a photon with incident angle 80° will have an internal angle of $\sim 20^o$ in bulk.

$$W_p(\lambda, \theta_3; dp^*, X_{PN}) = e^{-\mu \cdot dp^*/\cos\theta_3}(1 - e^{-\mu(X_{PN} - dp^*))/\cos\theta_3}) \tag{12}$$

$$W_n(\lambda, \theta_3; X_{PN}, dw^*) = e^{-\mu \cdot X_{PN}/\cos\theta_3}(1 - e^{-\mu(dw^* - X_{PN})/\cos\theta_3}) \tag{13}$$

Building upon the study of avalanche triggering probabilities [18], the internal efficiency $iPDE$ is given by multiplying the absorption weights by the avalanche probability of their respective minority carriers, $iPDE = P_e \cdot W_p + P_h \cdot W_n$. This convenient multiplicative form arises from the assumption that electrons (holes) generated anywhere within the effective p-region (n-region) survive and are drifted to the high-field region about X_{PN} where avalanches are initiated[5]. The

[5] this region is where impact ionization of e^- or h^+ is above threshold

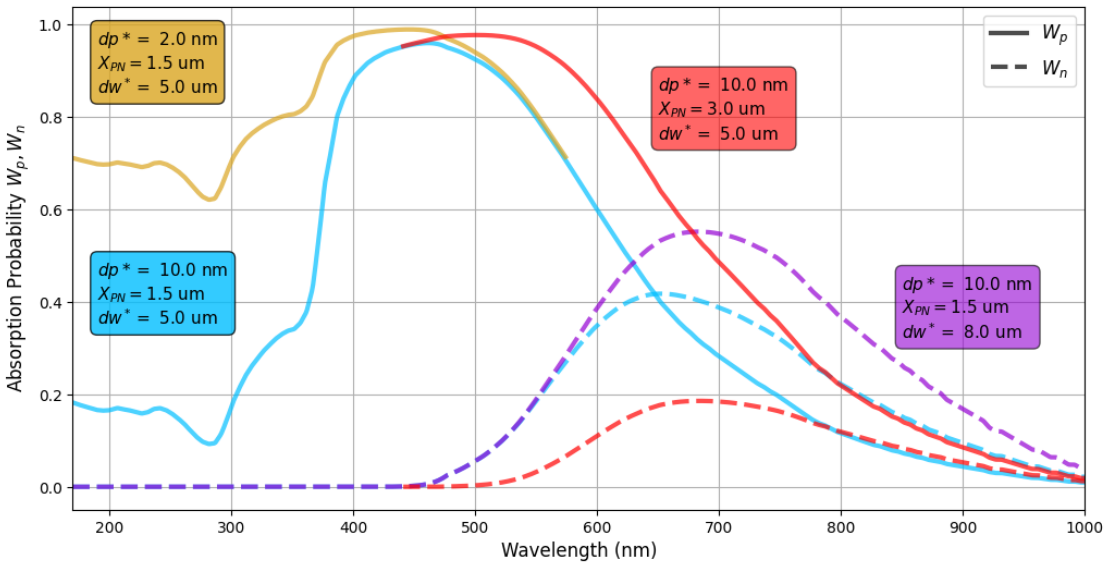


Figure 4: Absorption probability vs wavelength for the electron collection region W_p (solid lines) and hole collection region W_n (dashed). The different curves highlight the following influence of PN geometry: proximity to surface (dp^*) improves absorption in the UV, a deeper high-field region X_{PN} shifts the tradeoff between electron or hole driven avalanches, and deeper junction (dw^*) increases absorption for NIR wavelengths.

sharp transition from P_e to P_h about X_{PN} arises from the simplification that this high-field region is small relative to the photon absorption scale[6]. For describing avalanche efficiency we adopt the empirical parameterization $P_e(V) = A_e \cdot (1 - e^{-V/V_e})$ and similarly for P_h, as shown to describe data well in [16]. In the VUV regime where multiple e-h pairs can be generated the P_e term becomes a function of photon energy due to the quantum yield $\eta(\lambda) > 1$. Following from binominal statistics the electron-triggerred avalanche probability becomes $P_e(\lambda) \to 1 - (1 - P_e)^{\eta}(\lambda)$.

3.4. Fill factor and angular dependence

The fill factor FF represents the fraction of the SiPM's surface that is sensitive to incident photons. Design elements such as the quenching resistors surrounding each SPAD contribute to the insensitive fraction $1 - FF$. For most applications, FF may be treated as a fixed value independent of angle of incidence (AOI) or wavelength. However for large resistor heights relative to SPAD size the resistor may shadow the active area, effectively reducing FF for oblique angles of incidence. Simplifying the geometry, we treat the insensitive region as a symmetric, raised square loop surrounding the SPAD. With p the known SPAD pitch, we can calculate the one dimensional resistor width ff from the ratio of sensitive area to total area for a single SPAD, $FF = \frac{(p-2ff)^2}{p^2}$.

For incident light in a plane perpendicular to the resistor and assuming the resistor material is not reflective, the shadowed area is calculated. For a barrier of height h, with light at angle of incidence θ, the shadow length $h \tan\theta$ is multiplied by the sensitive region's side length $p - 2ff$. Defining the relative resistor width $W \equiv ff/p$ and height $H \equiv h/p$, the shadowed fill factor can be written as below, with F_o the fill factor at normal incidence. This model is fit to the angular

[6] Physically there is a small volume where photo-generated e/h pairs have an avalanche probability somewhere between P_e and P_h, as *both* charge carriers pass through a fraction of their respective avalanche inducing regions.

PDE data for the Hamamatsu VUV4 and compared to AFM measurements of resistor height in section 4. This effect is not significant in the FBK device.

$$FF(\theta) = F_o - \frac{(p - 2ff)h}{p^2}\tan\theta = F_o - (1-2W)H\tan\theta \tag{14}$$

3.5. Temperature and Photoabsorption

Operating temperature can affect the PDE via different mechanisms; modified optics, photoabsorption, carrier mobility and carrier freezeout. Carrier freezeout is neglected as it occurs around ~ 100 K, well below the measurement range of this paper [29]. We extend the model by including the temperature dependence of photo-absorption for bulk silicon. For indirect bandgap materials, photoabsorption decreases with temperature due to both the decrease in phonon population and the increase in the bandgap energy. As silicon's two lowest bandgaps (1.15, 2.25 eV) are indirect, photoabsorption is strongly temperature dependent in the NIR and weakly above the direct bandgap of 3.4 eV or ~ 380 nm. The absorption coefficient $\alpha(\lambda, T) \equiv 1/\mu$ was parameterized in silicon over 440-1000 nm from 4-300 K for applications in low mass DM searches [30]. We replace the direct bandgap description in [30] (taken from [31], [32]) with that from [33] as it appears more consistent with photoabsorption values calculated from k data.

Transforming this parameterized α to k using equation 11, the resonance about 360 nm is not present. Because of this lack of structure, and the systematically low attenuation lengths and sparse wavelength measurements in [30], we use the parameterization to calculate the relative change to $\alpha(\lambda, T)$ with respect to room temperature. Numerical smoothing is done about the direct bandgap threshold where a discontinuity arises[7]. This scaling is applied to the $\mu(\lambda; k)$ values calculated for the room temperature k data of the default dataset, yielding a more accurate temperature parameterization for photoabsorption in silicon, Equation 15.

$$\mu(\lambda, T; k) = \mu_o(\lambda; k)\frac{\alpha(\lambda, T = 300\text{K})}{\alpha(\lambda, T)} \tag{15}$$

4. Experimental Results and Model Fitting

This section describes the methods and results for fitting the model to the HPK and FBK PDE data. Fits are repeated for the default and Franta input optical datasets, with the temperature dependent photoabsorption correction (subsection 3.5) applied to the default dataset. All absolute PDE measurements are performed above 350 nm where the quantum yield is unity [16].

4.1. Fitting method

Subsets of the experimental data can constrain the parameters of the PDE model in a sequential and fairly independent manner, relevant for device characterization with limited apparatus capabilities. The measurements regimes and the corresponding model components they constrain are listed in Table 1 and detailed below, with results of the **sequential** or segmented fits refined via a global fitting routine.

Optical transmission into the device must be constrained first. The thicker oxide layer of the FBK device permits t_{ox} to be constrained using oscillations in the angular PDE (see subsection 4.6).

[7] The [33] functional form also reduces the discontinuity about this point relative to the others

Table 1: Summary of fitting procedures used to constrain different parameters/regions of the PDE model. For thin oxide, reflectivity vs angle is less powerful than vs wavelength.

Order	Physics	Measurement	Equation (PDE or other)	Constrained parameter
1	Optics (thin oxide)	UV reflectivity; λ	$R(\lambda; t_{ox})$, Equation 9	t_{ox}
1	Optics (thick oxide)	Relative $PDE(\theta)$	$PDE(\theta) \approx PDE(0)\ \frac{T(\theta)}{T(0)}$	t_{ox}
2	p depth	UV $PDE(\lambda)$ $\lambda \lesssim 420$ nm, high V	$FF \cdot T \cdot \mathbf{W_p}$	d_p^*
3	p depth and/or e- probability	UV $PDE(\lambda, V)$ $\lambda \lesssim 420$ nm	$FF \cdot T \cdot \mathbf{P_e} \cdot \mathbf{W_p}$	$P_e(V)$ and/or dp^*
3	e- probability (PDE independent)	SPE rate (V) $\lambda \approx 380$ nm	[16]	$P_e(V)$
4	Junction geo, h+ probability	Vis-NIR $PDE(\lambda, V)$	$FF \cdot T \cdot (P_e \mathbf{W_p} + \mathbf{P_h} \cdot \mathbf{W_n})$	X_{PN}, dw^*, $P_h(V)$

The Hamamatsu device has a thin oxide layer ($\lesssim$100 nm), making t_{ox} difficult to constrain with our data. Therefore the VUV reflectivity measured in [27] is fit as described in subsection 4.2. With the oxide thickness evaluated, the effective front of the junction $dp*$ and the voltage-dependent electron driven avalanche triggering probability $P_e(V)$ are constrained using short wavelength PDE. Measurements of the UV PDE at high overvoltage, assuming that $P_e \approx 1.0$, provides sensitivity to dp^* while measurements for multiple voltages constrain both variables if fit simultaneously. With $P_e(V)$ and dp^* constrained, visible and IR data probe the remaining junction parameters X_{PN} and dw^* with voltage dependence determining the hole-driven avalanche probability $P_h(V)$. Fitting in this segmented manner is referred to as the **sequential** method. Results of this method inform the initial guesses for the **global** fits where all parameters are floated simultaneously over all data. Alternatively, the simpler, PDE-independent measurement of P_e (see subsection 2.3) can be used as an input, however we compare the different P_e results to evaluate the model's accuracy.

For all voltage dependent data, the probabilities P_e or P_h are fit via two methods. In the **'parameterized'** method the shape parameters $V_{e,h}$ describing P_e, P_h are minimized simultaneously with junction parameters. The prefactor $A_{e,h}$ sets the saturation value, assumed unity although this assumption is tested explicitly. In the **'nested'** or unconstrained method the junction geometries are optimized globally for all voltages, while the probabilities P_e and P_h are treated as voltage-specific free parameters minimized for each voltage. The nested method introduces more degrees of freedom while not restricting the behaviour of $P_{e,h}(V)$ to any functional form. The combination of sequential/global, nested/parameterized and default/Franta input data, among other options, result in many fit results. The most informative cases are discussed below with the results summarized in Table 3 and Table 4.

4.2. Reflectivity and passivation layer thickness for the Hamamatsu VUV4

Fitting t_{ox} from our data is difficult as the VUV relative PDE has weak angular dependence and visible $PDE(\lambda)$ is convolved with $iPDE$. Stronger constraint is available from the VUV specular reflectivity data reported by the nEXO group[27], which we fit with equation Equation 9 to determine t_{ox}. Assuming only diffuse reflectivity from the insensitive silicon region, the expectation for specular reflectivity off the SiPM in the UV is $R_{SiPM} \approx FF_o \cdot R_{Si}$. The data in [27] is consistently lower than this by roughly 20-25%, but for a oxide thickness of approximately $\sim$ 17 nm the relative

shape agrees with the data. The data in [27] was fit for four different combinations of the the wavelength and angular-dependent reflectivity[8] for both devices. These fits report a ∼3 nm thicker oxide for a 75 um pitch device than a 50 um pitch device. Uncertain if this difference is by design or variation in the fabrication process, we take the weighted average of the 8 fits as the central value for t_{ox} and the maximum and minimum as the systematic bounds. The identical procedure is carried out using the Franta optical data, which shifts the central value from to 17.2 nm to 19.4 nm. Summary of t_{ox} values is given in Table 2.

Table 2: HPK oxide thicknesses derived from reflectivity data in [27], for given input optical dataset. First two columns give averages per device. Last column is weighted average and upper/lower limits.

Input data	**50 μm pitch**	**75 μm pitch**	**Estimated thickness (t_{ox})**
default	16.1 nm	18.8 nm	$16.2^{+2.4}_{-1.3}$ nm
Franta	17.3 nm	20.8 nm	$19.5^{+1.9}_{-2.7}$ nm

4.3. Wavelength-dependent PDE for the Hamamatsu VUV4

The absolute efficiency of the HPK VUV4 device was measured at 163 K from 350-830 nm for overvoltages 1-8 V. Experimental data with model fits, including P_e, are shown in Figure 5. Device parameters derived from the sequential and global fit methods are given in Table 3 and different P_h values are compared in the discussion. For the **parameterized** fits, the global fit agrees well with sequential fit using 420 nm as the upper wavelength limit for the UV data, and the 380 nm limited fit gives a shallower dp^* value. A_e was allowed to float in a global fit resulting in $A_e = 0.99 \pm 0.01$, validating the assumption that P_e saturates to unity.

The sequential nested fit (not shown in table) improves fit quality in the UV region, with the caveat that the 8 V P_e value is significantly below unity ($P_e(8\text{V}) \sim 0.93$), which we believe is physically inaccurate [16]. Due to this reduced P_e values, the Vis-NIR PDE in the sequential nested fit is underestimated where $W_n, W_p > 0$, resulting in overall poor fit quality. The nested global fit performs better, predicting $P_e(8\text{V})$ close to unity, a deeper dp^* and slightly deeper dw^*. Compared to the nested method, the parameterized fit results in a junction roughly $\sim 2\mu$m larger and yields lower P_h values. The global nested fit has a slightly lower reduced chi-squared. The nested global fit yielding $P_e(8\text{V}) \approx 1.0$ further validates the assumption that P_e saturates to unity.

Fitting using the Franta dataset significantly reduces fit quality, and results in different device geometry with a smaller junction size and increased P_h values. This is discussed further in section 5. Fits were repeated for the systematic uncertainty on the oxide thickness. For the global, parameterized fit using default optical dataset, all parameters are modified by less than 5% with the exception of dp^* by -13%, +43% for the 15.85 nm and 19.59 nm oxide values, respectively. The larger systematic impact on the VUV4's dp^* value is driven by the model's excess curvature in the UV. The relative changes are included in Table 3, with all other fits similarly impacted.

4.4. Model validation and extrapolation for Hamamatsu VUV4

The model was validated by extrapolating in wavelength down to 120 nm and comparing with the Hamamatsu reported PDE. Figure 6 shows the HPK values, our measured data at 4 V

[8] The shadowing effect in Equation 14 was included in the angular fit, with an extra factor of two added front of $\tan\theta$ as the specularly reflected light is blocked on entry and exit.

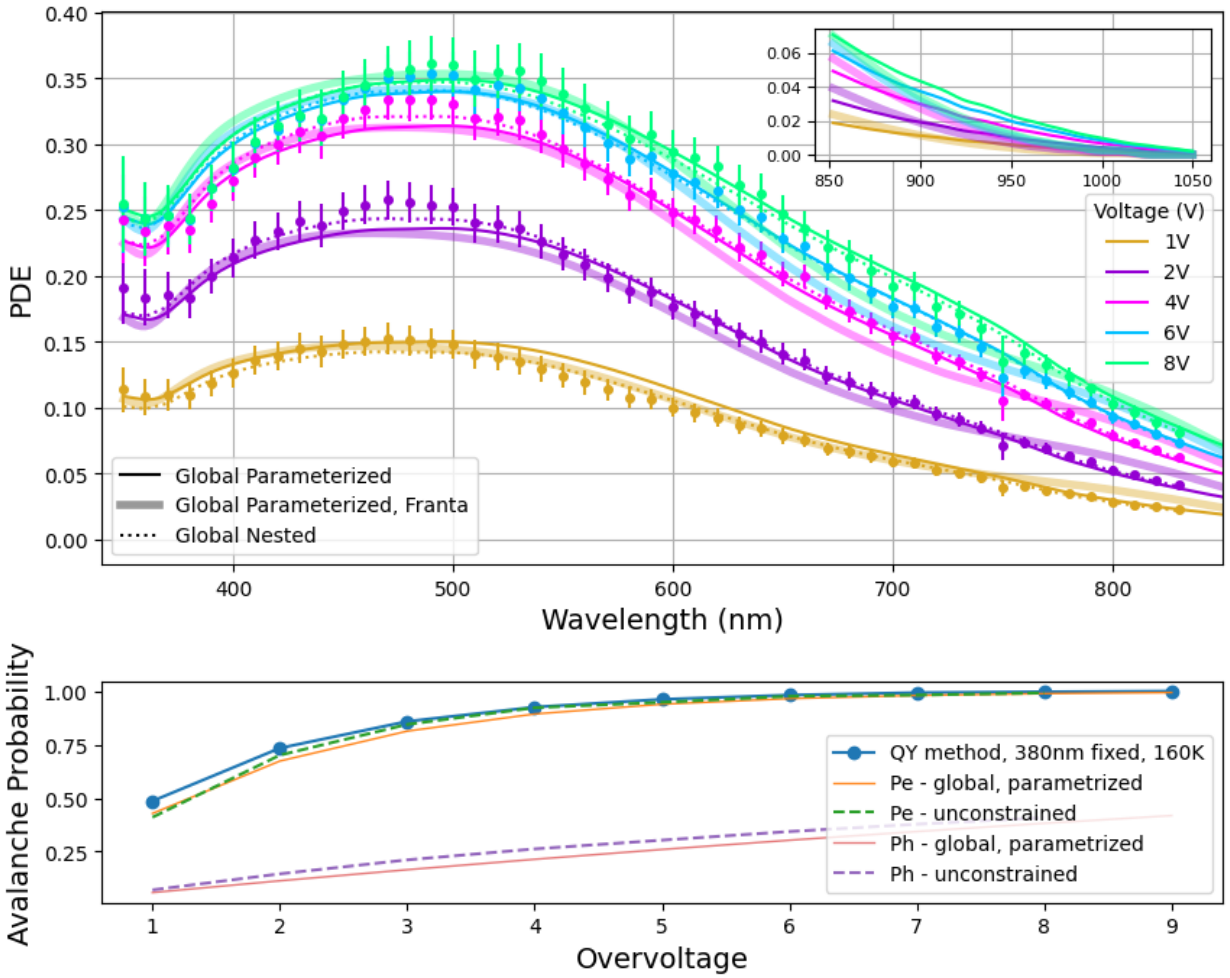


Figure 5: The HPK PDE for selected overvoltages is shown, with 3 fits overlayed. All fits perform similarly well wrt chi-squared. The 'nested' (unconstrained) P_e, P_h fits perform slightly better near the ∼500 nm maximum. At higher overvoltages the Franta fit (thick line) underestimate the PDE in the NIR. Inset shows both global parameterized fits extended to 1050 nm, where the Franta PDE drops slightly faster. The avalanche probabilities P_e and P_h are shown on the lower plot.

Table 3: A table showing HPK fit results. Where parameters overlap with those from the global parameterized fit, we insert a '-' for simplicity. For clarity we only include the oxide thickness systematic impact on the default, global parameterized fit; it's relative impact on the other fits is similar.

Fit technique	t_{ox} (nm)	dp^* (nm)	X_{PN} (um)	dw^* (um)	V_e or P_e(8 V)	V_h or P_h(8 V)	χ^2/ν
sequential (380)	default, $17.20^{+2.43}_{-1.31}$	0.67 ± 0.32	-	-	-	-	$^{193}/_{387}$
sequential (420)		-	-	-	-	-	$^{195}/_{387}$
global nested		1.31 ± 0.26	2.08 ± 0.08	12.4 ± 0.6	0.99 ± 0.01	0.41 ± 0.01	$^{107}/_{373}$
global parameterized		1.03 ± 0.24 $+37\%, -19\%$	2.35 ± 0.070 $+1.7\%, -0.8\%$	13.6 ± 0.8 $+3.7\%, -1.7\%$	1.80 ± 0.022 V $+1.7\%, -1.1\%$	16.76 ± 1.21 V $+5.1\%, -2.4\%$	$^{188}/_{387}$
global parameterized	Franta, $19.42^{+1.93}_{-2.7}$	0.82 ± 0.21	2.65 ± 0.13	5.8 ± 0.1	1.82 ± 0.02	4.68 ± 0.33	$^{705}/_{387}$

overvoltage and the model's prediction for both optical datasets and two temperatures. Beyond 500 nm, changing the model's temperature from 160 K to 273 K increases the predicted PDE to overlap with the datasheet, giving confidence to the temperature dependence implementation. To extend into the VUV, the quantum yield of silicon measured in [16] is applied to P_e as outlined in subsection 3.3[9]. The PDE model predicts the HPK reported VUV efficiency well for the default optical dataset. Using the Franta fit results unsurprisingly overshoots in the VUV due to that fit's shallower dp^*, due to the dataset's sharper n, k resonance. The default dataset's result is good evidence for the predictive power of this model, showing that VUV efficiency can be predicted using data for UV/IR efficiency, quantum yield, and VUV reflectivity (to constrain the oxide thickness). Extrapolation to wavelengths relevant for external crosstalk are shown in the inset of figure Figure 5.

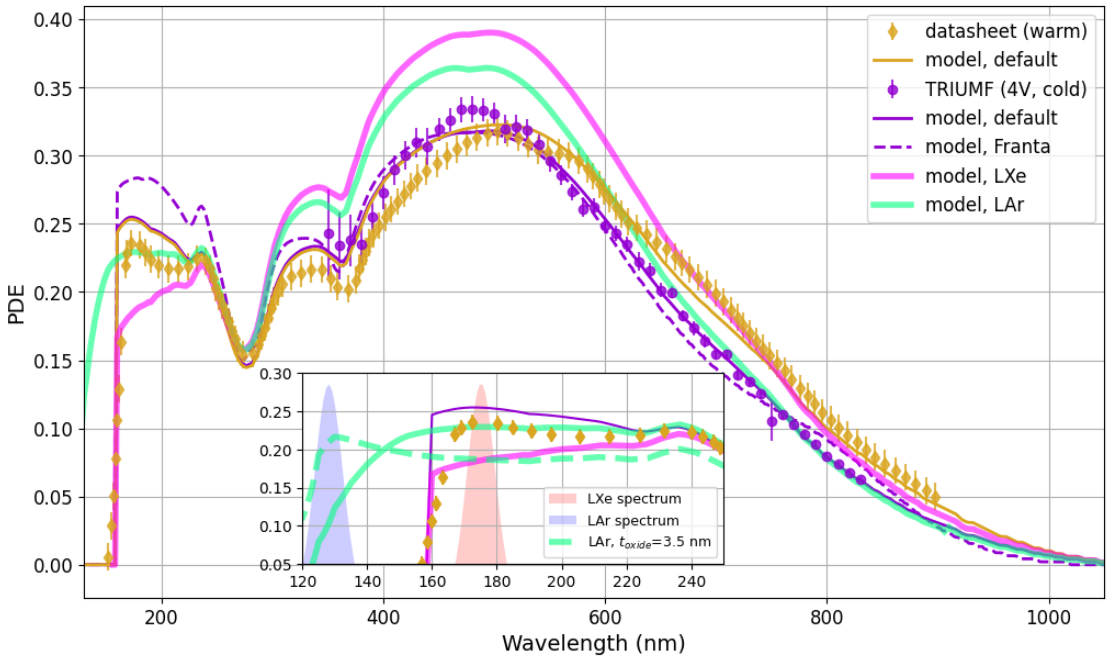


Figure 6: Comparison of the HPK reported PDE (warm), measured PDE (cold) and the model. Increasing the model's temperature from 160 K to 273 K brings the prediction closer to the datasheet in the NIR. The Franta fit overestimate's the PDE in the VUV due to shallower dp^*. The thicker lines are the estimated PDE in LXe, and LAr without quartz window. Inset shows the VUV region in detail, and another LAr PDE curve with a very thin oxide, predicted to yield higher PDE (see subsection 5.5).

The model's prediction for PDE in liquid xenon (LXe) and argon (LAr) are also given in Figure 6. The increase in the visible region is due to improved coupling between the SiO_2 and the active medium. The LXe transmission decreases below 250 nm as $n_{LXe} \approx n_{SiO_2} < n_{Si}$ and the enhancement from constructive interference is greatly reduced. This reduction of efficiency in LXe is not inconsistent with other preliminary measurements [34]. In LXe the HPK and average FBK thicknesses from this work yield transmissions of 38% and 39%, respectively. According to equation Equation 6, 40% transmission is the maximum achievable value for $t_{ox} = 44$ nm. For Xe

[9] We parameterize the quantum yield roughly as $\eta(\lambda) = (\frac{300nm}{\lambda})^{3/2}$

scintillation in vacuum the HPK t_{ox} yields the maximum possible transmission of $\sim 55\%$, while for FBK it is 40%. As such, the HPK device's oxide thickness is well-optimized for detection of the 175 nm centered Xe emission in both vacuum and LXe.

The falloff in the PDE at 160 nm is due to absorption in the SiO_2 of the thick window. In [35] Pershing *et al.* measures a PDE of ≈17% for LAr scintillation on a VUV4 with its quartz window removed. Our default fit predicts a PDE in LAr below 10%, and 13% using Franta. However 17% is achievable for a thinner oxide layer, possible given the devices are from different fabrication batches. Lower oxide k values than those input would also explain the relatively high reported LAr-PDE. This highlights the importance of reliable k data for SiO_2 at VUV wavelengths, which may present a challenge given that the optical properties of SiO_2 are known to vary with growth conditions and techniques [36]. The sharp slope of the PDE in this vicinity (inset in Figure 6) also emphasizes the importance of accurately known emission spectra.

4.5. Angular dependence of PDE for the Hamamatsu VUV4 - effects of device microstructure and packaging

The angular dependence of PDE was measured for the HPK VUV4 device at select wavelengths between 190-830 nm. Within the experimentally achievable angles, the predicted transmission for $t_{ox} \approx 17$ nm has minimal, monotonically decreasing dependance on AOI. We observed a small discrepancy between the predicted and measured $PDE(\theta)$, assuming FF is fixed.

The angular data (subsection 2.2) is first normalized by the signal at normal incidence. The transmission is then factored out using the oxide thickness derived from the reflectivity analysis. This results in a 'measured' fill factor ($FF(\theta) \approx PDE(\theta)/T(\theta)$), as the internal PDE is independent of AOI. This effective fill factor is again normalized by $FF_o = 0.6$ and is shown in figure 7. The falloff at high AOI is due to beam containment and blocking by the device package, as onset varies with pinhole size. The normalized, transmission-corrected data is not flat. For UV wavelengths, this effective $FF(\theta)$ is decreasing with angle and for longer wavelengths the slope of $FF(\theta)$ becomes positive with respect to increasing AOI. The offset of $FF(\theta)/FF_o$ from unity at 25° is plotted versus wavelength, showing a monotonic wavelength dependence.

We assume for UV wavelengths the shadowing model (subsection 3.4) accurately describes the system, as the polysilicon material is highly absorbing and diffraction should be small[10]. We fit the normalized UV data (190-235 nm, solid lines in figure 7) from -15 to 30 degrees for an effective resistor height h using equation 14. The fit yields a value $h = 2.6 \pm 0.1$ um and a reduced chi-squared of 1.6, in excellent agreement with AFM measurements performed by the nEXO collaboration[11] which list the resistor height between 2.5 um and 3 um. Using the upper and lower values of t_{ox} or the Franta derived values all give acceptable resistor heights. For longer wavelengths reflection and diffraction are expected to increase and this model will no longer describe the physics, as suggested by the data where $FF(\theta) > FF(0°)$.

4.6. Angular dependence of PDE for the FBK VUV-HD3

The relative angular PDE of the FBK device was measured in three separate campaigns, see Figure 8. The second and third campaigns extend to -10°. A small correction to the angle's

[10] This was estimated by treating the system as the textbook 'half-plane diffraction' problem and integrating the total intensity of E field diffracted below the plane.

[11] Acknowledgements to Prabandha Nakarmi, who performed this measurement in 2022

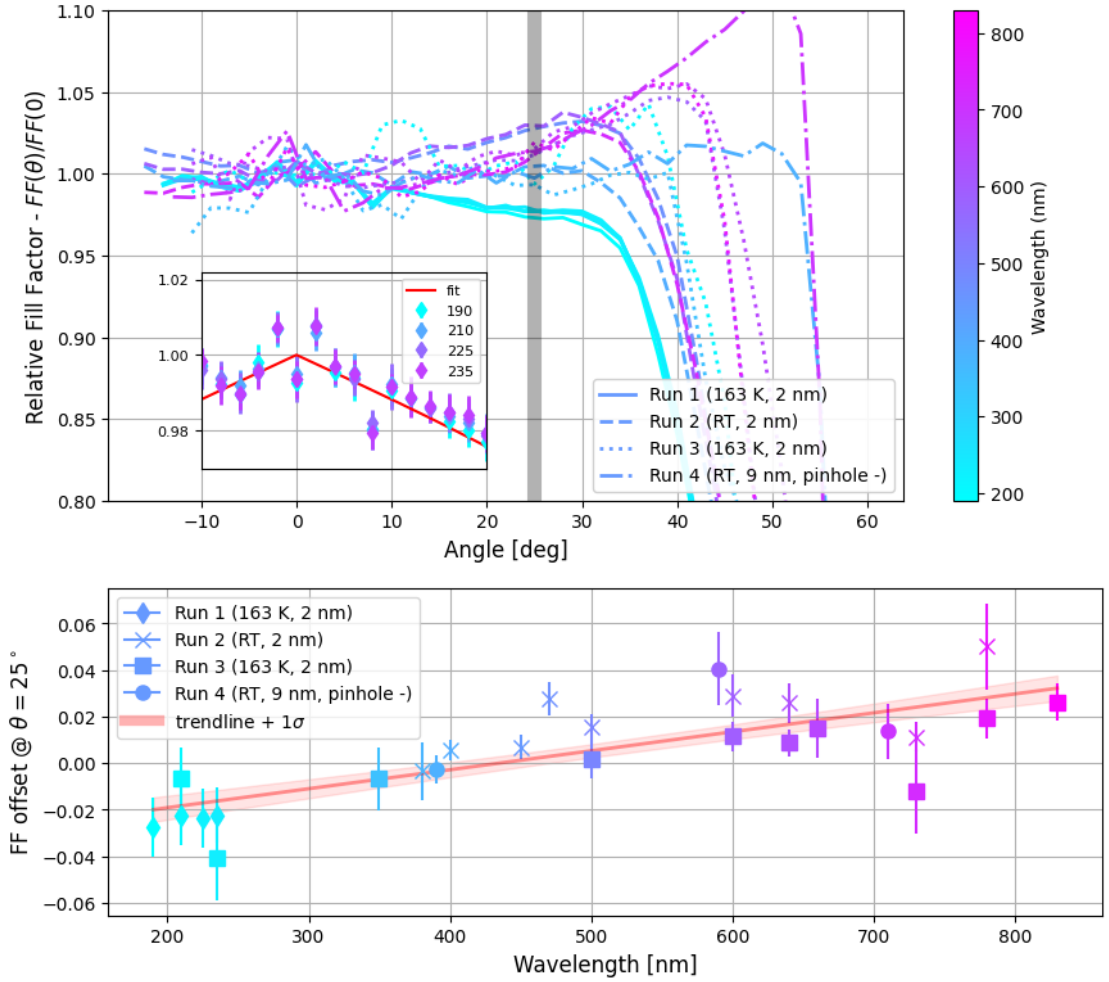


Figure 7: Upper plot showing $FF(\theta)/FF(0^\circ)$, derived from relative PDE data corrected for transmission. Inset shows only the UV data fit with the shadowing Equation 14 to find the effective resistor height. The lower plot shows the offset in FF wrt unity about 25deg (dark vertical line), versus wavelength. Some experimental data omitted from upper figure for clarity.

zero-point is calculated by assuming the data should be symmetric between ±10°, which resulted in a shift of 2.5 degrees in the positive direction and is applied to the relative transmission function fit to the data. Each campaign produces slightly different oscillation patterns. A variety of global fit combinations were attempted for a single oxide thickness and various experimental nuisance parameters (wavelength shifts, angular offsets and MC width) that depended on dataset or wavelength region. These efforts were unable to produce consistent results, and it was necessary to assign a different oxide thickness to each campaign. Although the smallest and largest oxide thicknesses are within 2% of the mean value of 1358 nm, they are not within fitting uncertainty and the observable oscillation peaks and valleys are clearly inconsistent. The resulting thicknesses using the default or Franta datasets are within 0.5 nm of one another.

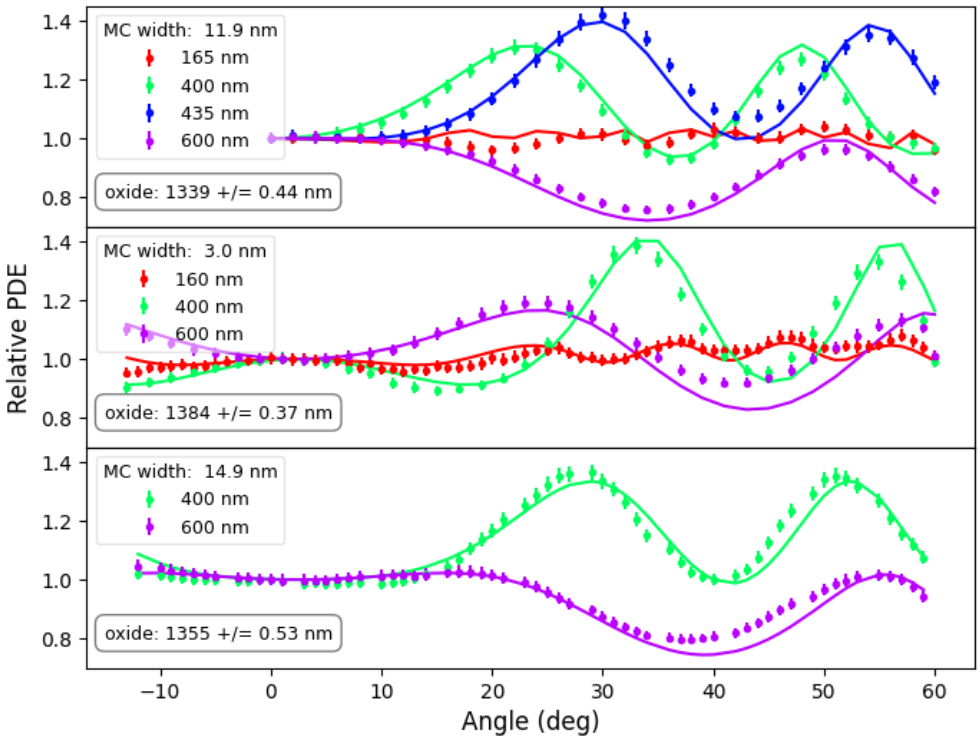


Figure 8: Three measurements of FBK device's relative PDE versus angle, fit to extract the oxide thickness. Monochromator FWHM (MC width) are noted in the legends, and fitted oxide thicknesses displayed in the lower left. The UV data (red) are shifted by an additional wavelength offset of -10 nm (upper) and -15 nm (middle), which is attributed to the spectral non-uniformity of the light source.

The 160, 165 nm UV data are not included in the fits as we suspect a large spectral uncertainty due to the underlying peak in the deuterium lamp's spectrum. In Figure 8 the UV wavelengths of 160 and 165 nm are treated as floating parameters in the fit and shifted by -10 and 15 nm, respectively. The MC slit width is also floated to provide further flexibility. This is intended to demonstrate it is possible to fit the data under reasonable assumptions. No shadowing was observed for the FBK devices, likely due to smaller or asymmetric resistors.

4.7. Wavelength-dependent PDE for the FBK VUV-HD3

PDE for the FBK device was measured from 350-830 nm at 1-6V overvoltage. Accurate measurement at higher overvoltage was not possible due to an increased rate of correlated avalanches confounding pulse-finding analysis. Data are shown with fits in figure Figure 9. The sequential fitting method described in subsection 4.1 was applied to the data with some fit results shown in table Table 4. The oxide thickness was derived from the wavelength measurements giving 1358.6 nm, which is in agreement with the average from the angular measurements. This thickness was fixed for the sequential fits.

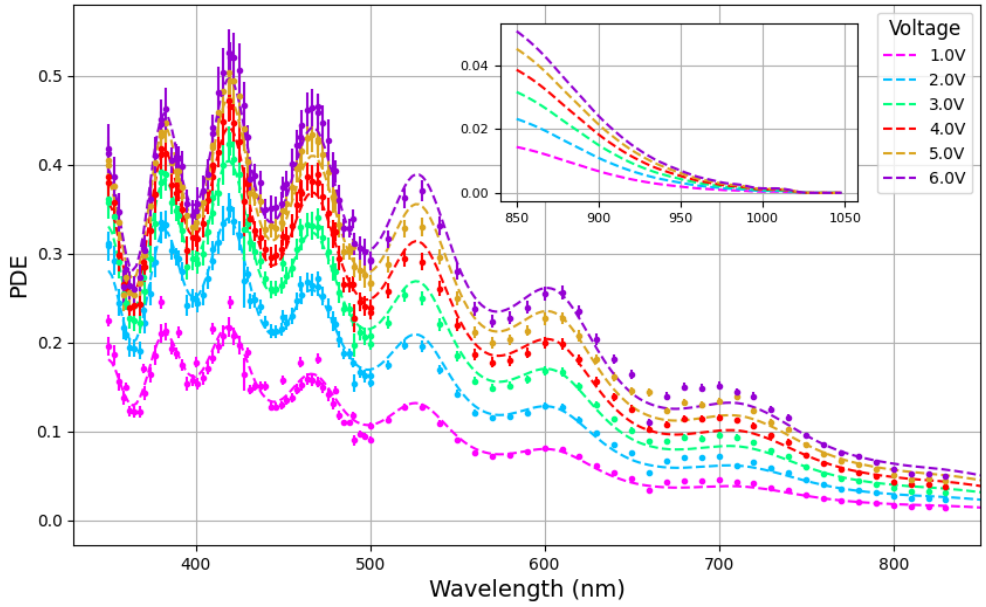


Figure 9: FBK data and the global, parameterized fit result using the default input dataset. Other fits not shown for clarity. Can see some outlying data points in 1 V curve from the lower resolution wavelength scan. Inset plot shows PDE extrapolated to 1050 nm. The FBK P_e curves are almost identical to those shown in Figure 5.

Table 4: A table showing FBK fit results from PDE vs wlen. Where parameters overlap with those from the global parameterized fit, we insert a '-' for simplicity. ** denotes the oxide thickness is fixed to known best value.

Fit technique	t_{ox} (nm)	dp^* (nm)	X_{PN} (nm)	dw^* (um)	V_e or P_e(6 V)	V_h or P_h(6 V)	χ^2/ν
sequential (380 nm)	1358.9**	1.62 ± 0.13	420 ± 2.6	7.1 ± 0.1	1.56 ± 0.02	15.8 ± 0.2	2037/589
sequential (420 nm)		-	-	-	1.81 ± 0.01	-	1736/589
global nested	1358.9 ± 0.2	1.12 ± 0.08	498 ± 5	8.43 ± 0.13	1.01 ± 0.004	0.26 ± 0.002	1528/578
global parameterized	1358.9 ± 0.2	0.90 ± 0.06	513 ± 5	8.39 ± 0.13	1.83 ± 0.01	18.90 ± 0.27	1734/588
global param, Franta	1359.7 ± 0.4	0.58 ± 0.05	659 ± 14.4	3.82 ± 0.06	1.78 ± 0.013	8.56 ± 0.23	2359/588

For the **parameterized** fits, the sequential and global fits gave parameters outside uncertainties for a UV fit limit of 380 nm. The two methods agree if the UV limit is increased to 420 nm. Similarly to the Hamamatsu device, the **nested** sequential fits reduce dp^* and P_e to reduce the model's curvature in the UV, which then underestimates the visible PDE. The global nested fit

predicts similar junction geometry to the parameterized fit, slightly improving the reduced chi-squared. As for HPK, repeating the process with the Franta input dataset yields a much smaller junction and higher P_h values. The junction geometries for both global fits are not quite within uncertainties of one another. The inset of Figure 9 shows the model extrapolated to wavelengths relevant for external crosstalk.

5. Discussion

We have successfully demonstrated that the many parameters of the PDE model can be independently constrained using combinations of relative optical measurements and absolute PDE from the UV to NIR. The model's flexibility allows it to accurately describe measured PDE for two different optical datasets, and has shown predictive power to extrapolate the PDE for new wavelengths, temperatures and operational media. For improvement in accuracy, the uncertainty with respect to input optical data requires further investigation.

The model predicts excess curvature about 360 nm which drives much of the model's imperfections. As simultaneously decreasing dp^* and P_e flattens the model's UV response, fineness is required to match the sequential and global fits. This highlights the need for absolute PDE measurements below 350 nm to better assess dp^* separate from P_e. The Franta dataset's sharper n, k resonance further encourages a shallower junction, driving the overestimated VUV PDE for Franta (see Figure 6). For the wavelengths available in this study the parameterization of P_e, P_h is preferred as the unconstrained fits provide excess degrees of freedom, and in addition some unconstrained Franta fits yield nonphysical P_h values. For UV limited setups, the rate-based P_e measurement can be input. Future work may investigate a model with probabilistic charge collection about dp^*, which preliminary modeling showed was effective at flattening the response in the UV while retaining $P_e = 1.0$.

5.1. Sensitivity to input optical data

Compared to the default dataset, fitting with the cryogenic, lower k Franta dataset retains an acceptable quality of fit while yielding a much smaller junction and correspondingly higher P_h. This is due to Franta's decreased photoabsorption and different shape in the Vis-NIR region. This highlights the model's degeneracy for certain k values and dw^*, P_h resulting in a systematic uncertainty in regards to which dataset and corresponding fit results are more physical. The interplay between photoabsorption and junction size, P_h is not straightforward. As SiPMs are highly doped devices, the increase in photoabsorption with doping concentration[37] suggests the higher k default dataset is more physical. Also considering the better quality of fit, we assume the default dataset results are more representative and recommend it's use in applications of this work. Future measurements should better constrain dw^* or P_h, possibly rejecting a set of results. Direct study of P_h could be achieved using a two-photon absorption technique [38] or by studying the rate of delayed crosstalk, which is expected to be entirely hole-driven in P-on-N devices. In addition, photoabsorption on impurities should be assessed due to its possible enhancement of ExT near the bandgap. Given a known doping profile, a rigorous approach would use the complete Franta model[23] to calculate n, k as a function of junction depth and numerically evaluating the W_p, W_n integrals.

For the liquid noble data, there is uncertainty in the exact value of n at long wavelengths ($\lambda \sim 1000$ nm), and if the IR resonance used for the Sellmeier parameterization in [24] is physical and expected. For LXe our estimated range of values between $n(1000\text{nm}) = 1.35-1.4$ only modifies

the $\sim 80\%$ transmission into silicon by 1.2%[12]. If ExT simulations with more precision are needed the exact value of n in the IR should be addressed.

5.2. *Avalanche triggering probabilities*

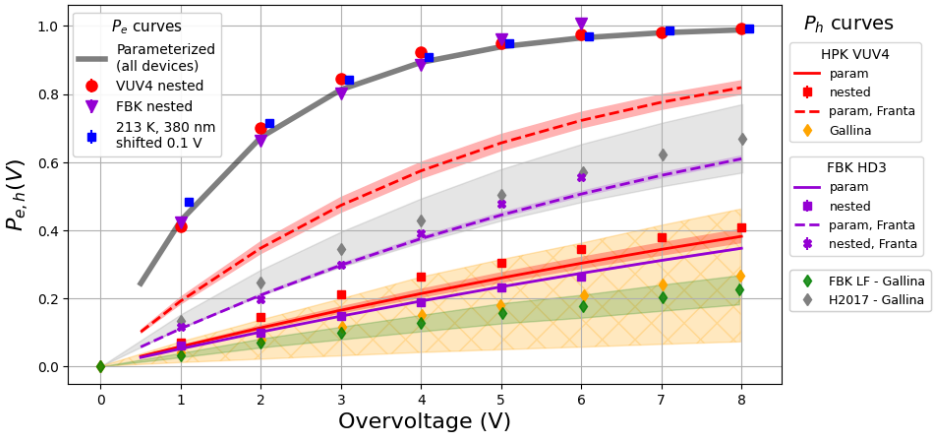


Figure 10: Comparison of avalanche probabilities. Upper curves are P_e from the fits and direct measurement (blue squares), all global parameterized fits give essentially the same P_e curve (black line). Lower P_h curves are the global fits from this work and three P_h curves from Gallina [18]. The Franta driven P_h values are higher than the default dataset's due to the lower photoabsorption in the NIR, resulting in a systematic uncertainty on P_h. Other than the Franta VUV4 fit, the P_h curves fall within the range of values taken from [18].

Figure 10 compares P_h and P_e curves derived from the PDE fits, PDE-independent measurements (subsection 2.3), and those reported by Gallina [18]. The VUV4, unconstrained Franta fit is omitted due to nonphysical $P_h > 1$. The similarity of the P_e curves is remarkable, despite coming from different devices, optical data, measurement techniques and temperatures. The assumption that P_e saturates to unity was validated both by the unconstrained fits and the parameterized fits where A_e was varied. All global fits yield P_e close to a parameterized curve with shape parameter $V_e \approx 1.8$ V. The raw rate-based P_e measurement for the VUV4 device (subsection 2.3, blue squares in Figure 10) has similar curvature but is offset from the fit values. Allowing for a horizontal shift in voltage of 0.1 V, owing to uncertainty in breakdown voltage calculation between different datasets, the curves are more similar. Further work may test the universality of P_e between devices, as values in [18] are also extremely similar to those measured here.

Compared to P_e, the extracted P_h curves have large variance. For both devices, the default-dataset derived P_h values are low and fall within the large uncertainty bands of the VUV4 device measured in [18]. The Franta derived values are almost twice as high and are comparable to the results for the so-called H2017 device from [18]. The parameterized versus unconstrained fits are in better agreement for the FBK device, possibly due to better optics constraints from the oscillations. The large variance in P_h across different devices is not unexpected for moderate overvoltages, as the lower mobility of holes prevents P_h from quickly saturating. The assumption of a sharp transition from P_e to P_h at X_{PN} may also contribute to the spread, as larger junctions may have a less-well defined singular value for P_h. The sharp-edge assumption may be tested in the future.

[12] 1.35 taken from the pre-resonance region in [24], 1.4 from $n \approx \sqrt{\epsilon_r}$, $\epsilon_r = 1.96$ reported in [39].

5.3. Angular dependence of PDE

We consider that the variation in oxide thickness between FBK angular PDE measurements may be caused by variation in the surface layer, as the beam's exact position varied between installs. However the discrepancies in the data highlight the need for a more precise understanding of the experimental optics, namely the lamp's underlying spectrum and MC transfer function. As oscillations in equation Equation 6 are not symmetric with respect to θ or λ, the averaged transmission's maxima or minima can shift depending upon the transfer function. In the HPK device we measured shadowing due to the the surface structure, a small effect. We do not foresee this level precision impacting detector response and is outside the motivation of this model.

5.4. Temperature dependence of PDE

The third FBK angular campaign measured the relative PDE at 213 K and 163 K to test for temperature dependance in the thin-film interference. This produced identical oscillation patterns for both temperatures, indicating that the surface optics, at least at the measured wavelengths of 400 and 600 nm, are unperturbed by temperature in that regime. The parameterized temperature dependence of the photo-absorption appears reliable, although in the future we hope validate it on PDE data below the 440 nm limit from [30]. As discussed in subsection 2.3, P_e showed negligible temperature dependence for the ranges measured; the change in mobility does not seem to strongly impact P_e. Carrier-freezeout is expected to contribute at colder temperatures and future work may include this effect.

5.5. Optimizing PDE for Liquid Nobles and Quantum Sensing

For specific applications in quantum computing and particle physics, we investigate the theoretical upper limit of PDE for P-on-N devices using the single film oxide-silicon surface model. This requires maximizing the fill factor FF, the optical transmission and achieving junction geometry such that avalanches are electron-driven ($W_p \to 1.0$). In this case the PDE is approximately $PDE_{max} \approx FF \cdot T_{max} \cdot W_p$, provided the devices can be operated at overvoltages such that $P_e \to 1.0$. This requires the correlated avalanches associated with higher overvoltages remain experimentally tractable. Maximizing the fill factor is a challenging engineering and fabrication challenge, driven by the need for a low field boundary around each SPAD, quenching resistors and isolating trenches. Backside-Illuminated (BSI) structures are a promising approach, reaching FF values close to 100% [40]. Multilayer anti-reflective coatings can further improve transmission in the UV [41] and achieve near-100% transmission for visible light [42, 43, 44]; however this comes with an increased cost, complexity and is outside the scope of our single thin-film optical model.

The upper half of Table 5 correspond particle physics applications using argon or xenon. The wavelengths listed in the lower half are relevant for qubit manipulation in quantum computing and free-space laser operation in satellite-to-ground quantum key distribution [11]. Optimal oxide thicknesses are calculated by maximizing Equation 6 at normal incidence. The listed junction geometries are calculated such that $W_p = 0.9$. The liquid noble applications include the effect of optical coupling with the medium and the PDE is convolved with the emission spectra[13]. We note for LXe scintillation, both HPK and FBK devices characterized in this work achieve ~90% internal PDE when operated at high overvoltage.

[13] this can modify the transmission at the percent level, especially as emission is close to medium's n resonance

Table 5: Optimized SiO_2 thickness and junction geometries for maximizing high PDE. Relevant geometries are given for achieving $W_p = 0.9$. For UV wavelengths (liquid nobles) the depth of the electron collection region dp^* is given. For longer wavelengths the junction center X_{PN} is given. Similar transmissions can be obtained for multiple oxide thicknesses, only values below 1 μm are reported.

Application	**λ (nm)**	**Maximum transmission**	**t_{ox} (nm)**	**dp^* or X_{PN}**
LAr in vacuum	127 ± 10	0.44	4.7	0.6 nm
LAr	127 ± 10	0.38	3.45	0.6 nm
Xe-doped LAr	175 ± 10	0.46	17.3	0.5 nm
LXe	175 ± 10	0.40	44	0.5 nm
TPB [45], PEN [46] in LAr	430 ± 80	0.76	71	$X_{PN} = 550$ nm
Diamond NV center [47]	532	0.89	91.3	1.7 μm
Sodium D-line	589	0.91	101	3.3 μm
NV center (zero-phonon) [48]	637	0.91	328	5.6 μm
Rubidium D1, D2 lines [49]	780, 795	0.93	410, 671	21, 27 μm
Satellite-Ground QKD [11]	850	0.93	731	45 μm

For Lar emission, the input SiO_2 optical data is absorbing thus a minimal oxide is preferred. As in subsection 4.4, this highlights the importance of accurate k data for SiO_2 at VUV wavelengths. The results suggest that, in the native media, it is possible to achieve PDE close to 35%. Notably, 6% higher PDE can be achieved for Xe-doped LAr than pure LXe due to the transmission enhancement possible via constructive interference. For the wavelengths relevant to quantum applications, a transmission of ∼90% is possible. Combined with $W_p \approx 0.9$ this yields an overall PDE of approximately 80%. We anticipate that future device designs can push FF close to unity while retaining high avalanche efficiency, timing resolution and suppressing correlated avalanches. Digital SiPMs may provide the best path forward for their insensitivity to correlated avalanches within a single SPAD [50]. This discussion of course ignores the complexity of silicon doping and the engineering of suitable electric field profiles.

6. Conclusion

We have successfully demonstrated that the many parameters of the PDE model can be independently constrained using combinations of relative optical measurements and absolute PDE in the UV-NIR. The PDE data for the Hamamatsu VUV4 and FBK VUV-HD was fit, giving junction parameters, avalanche probabilities and oxide thickness for each device. Compared to HPK, the FBK oxide is roughly 100 time larger and the junction almost half the size. All measured P_e values are similar, with a large spread in P_h values. We consider the default optical dataset results more physical, and are able to not only reproduce device PDE but extrapolate to operation in liquid nobles, from the VUV to NIR wavelengths relevant for external cross-talk. This work provides a foundation for improved detector simulation and further SiPM characterization and understanding.

The PDE model and lookup tables for simulation will be made available online.

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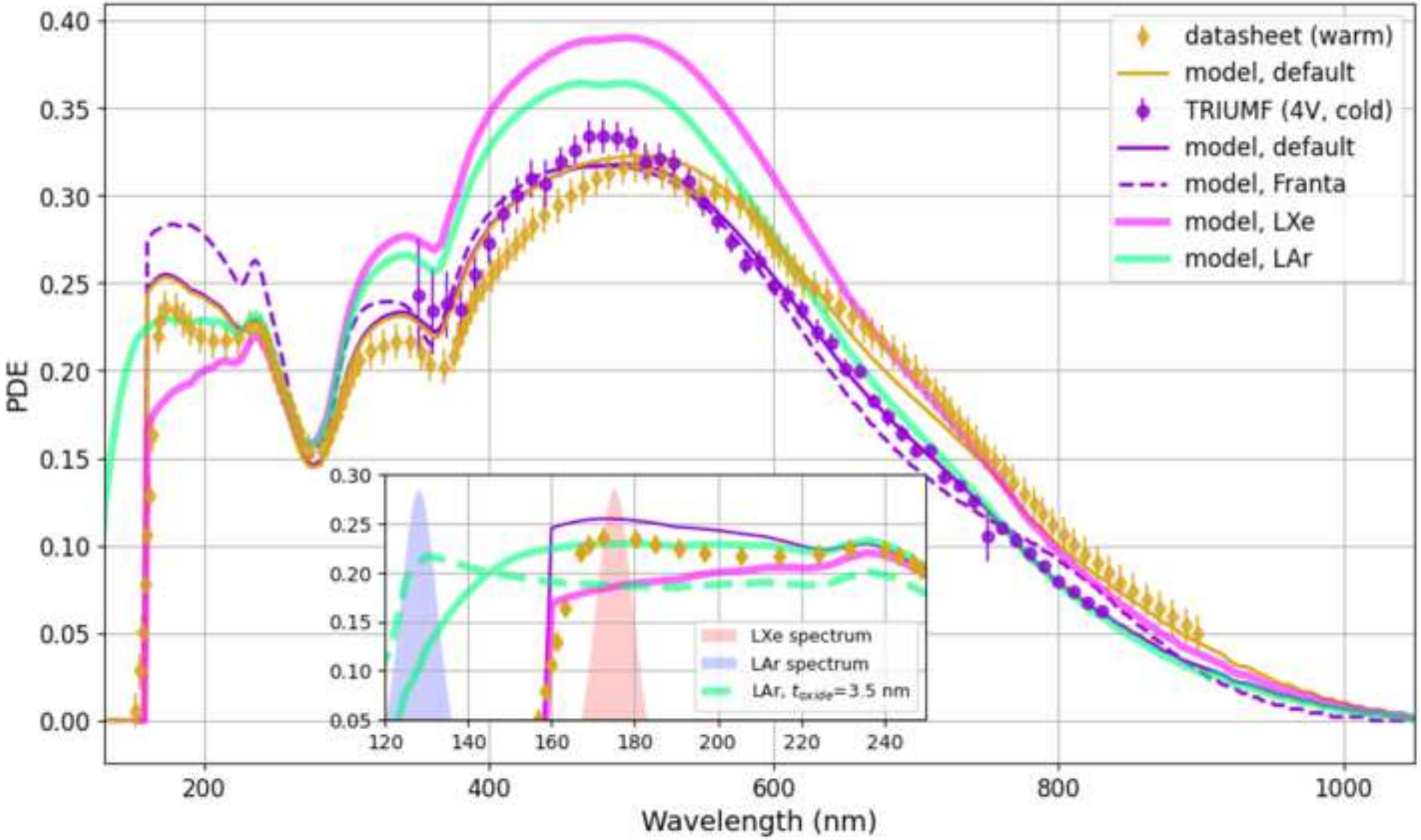
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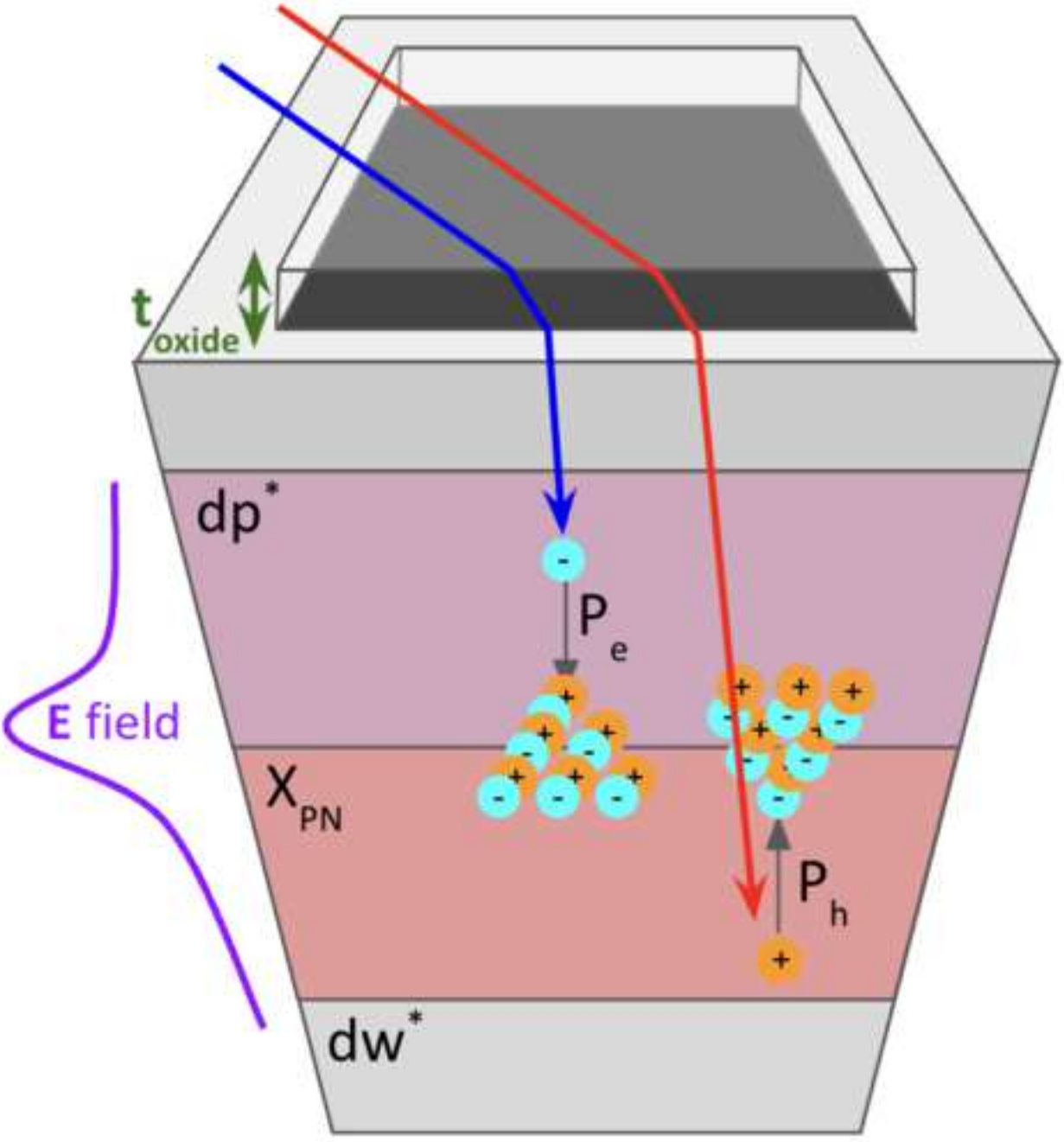
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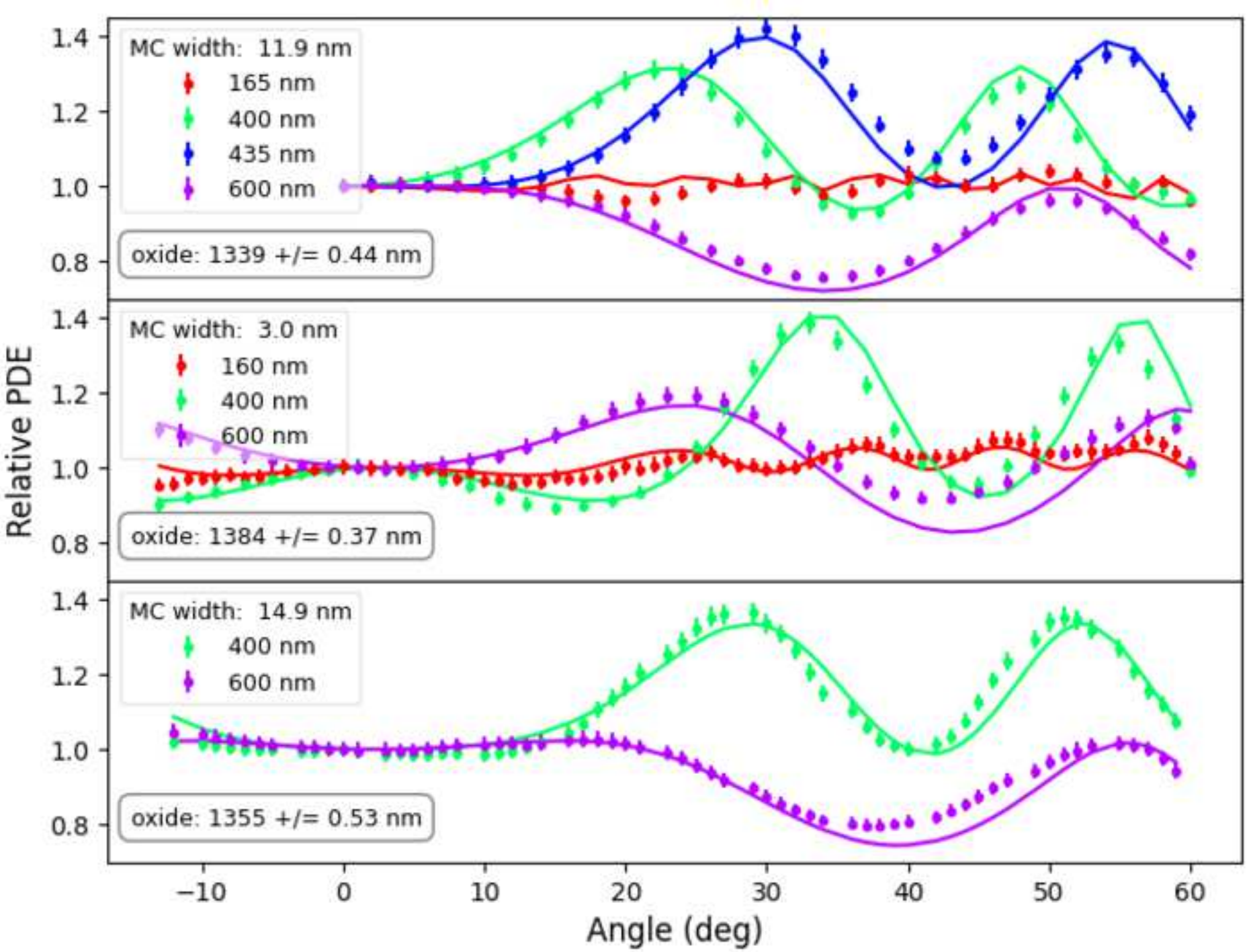
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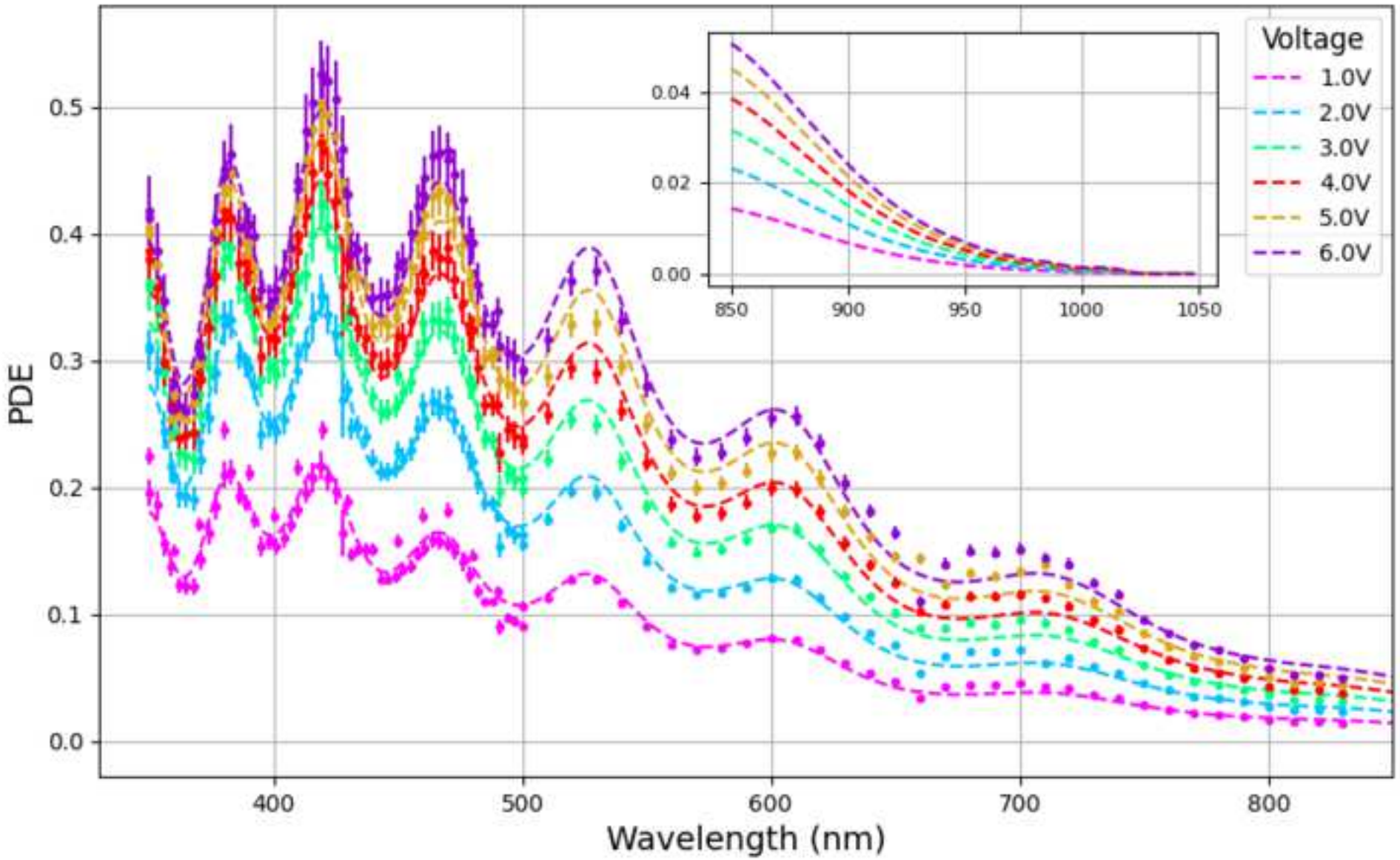
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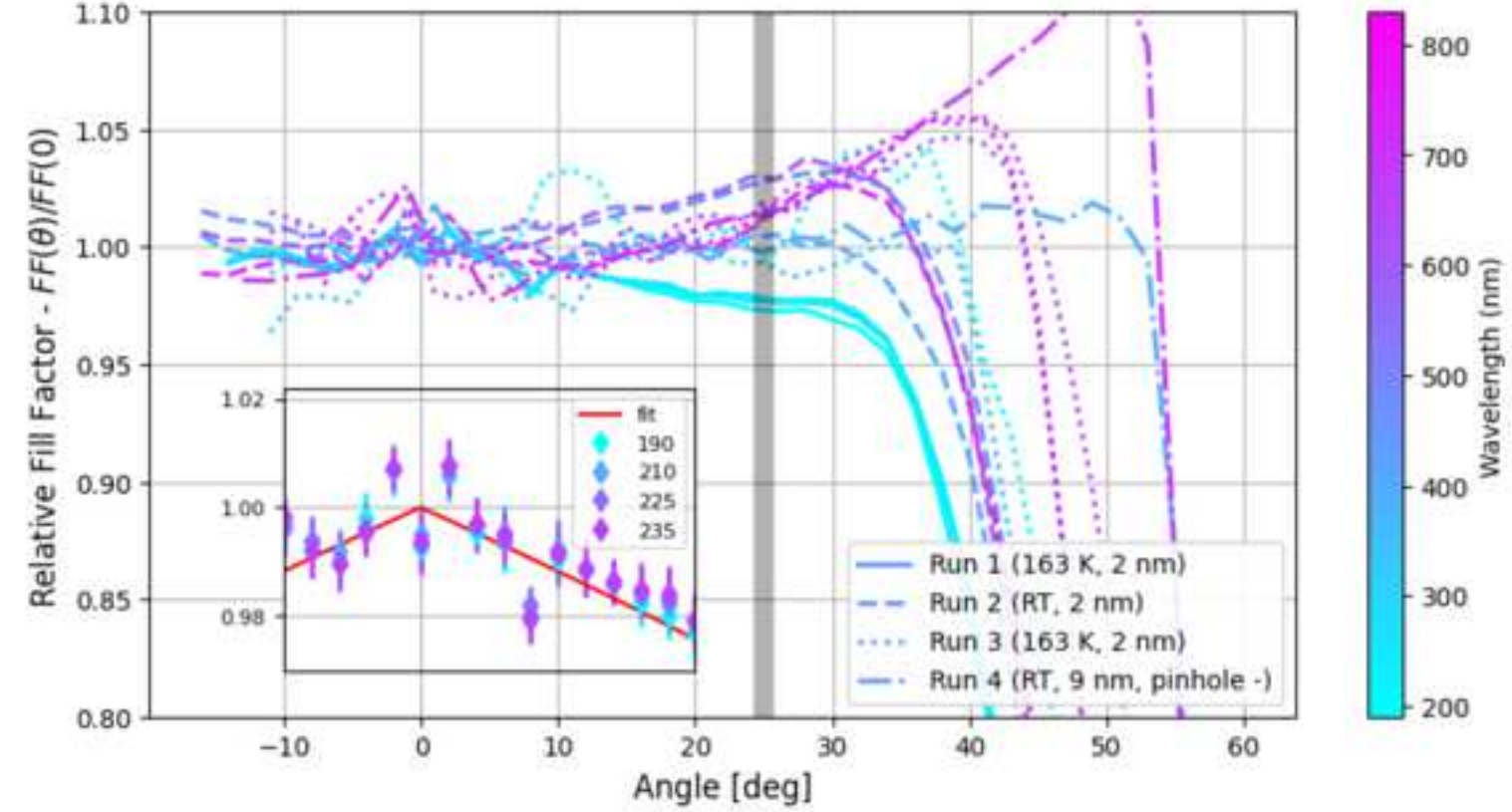
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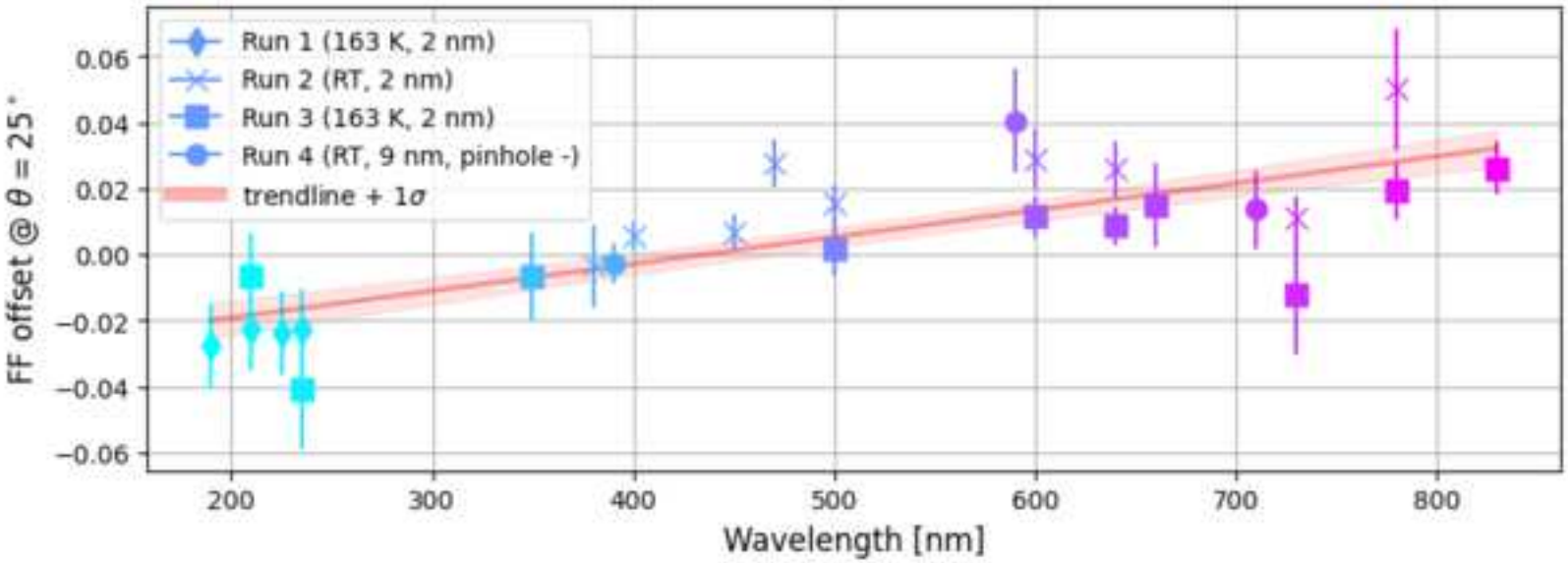


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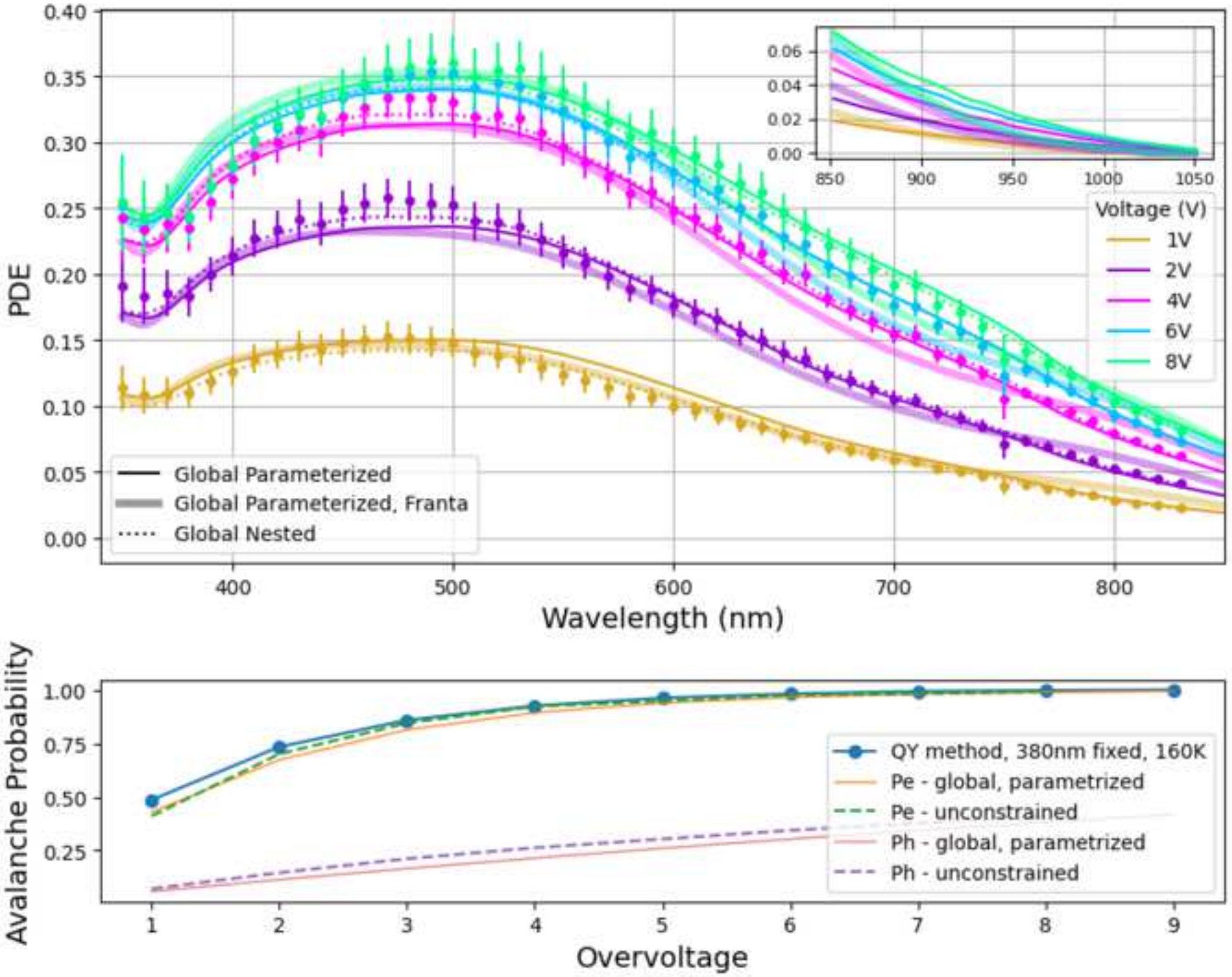


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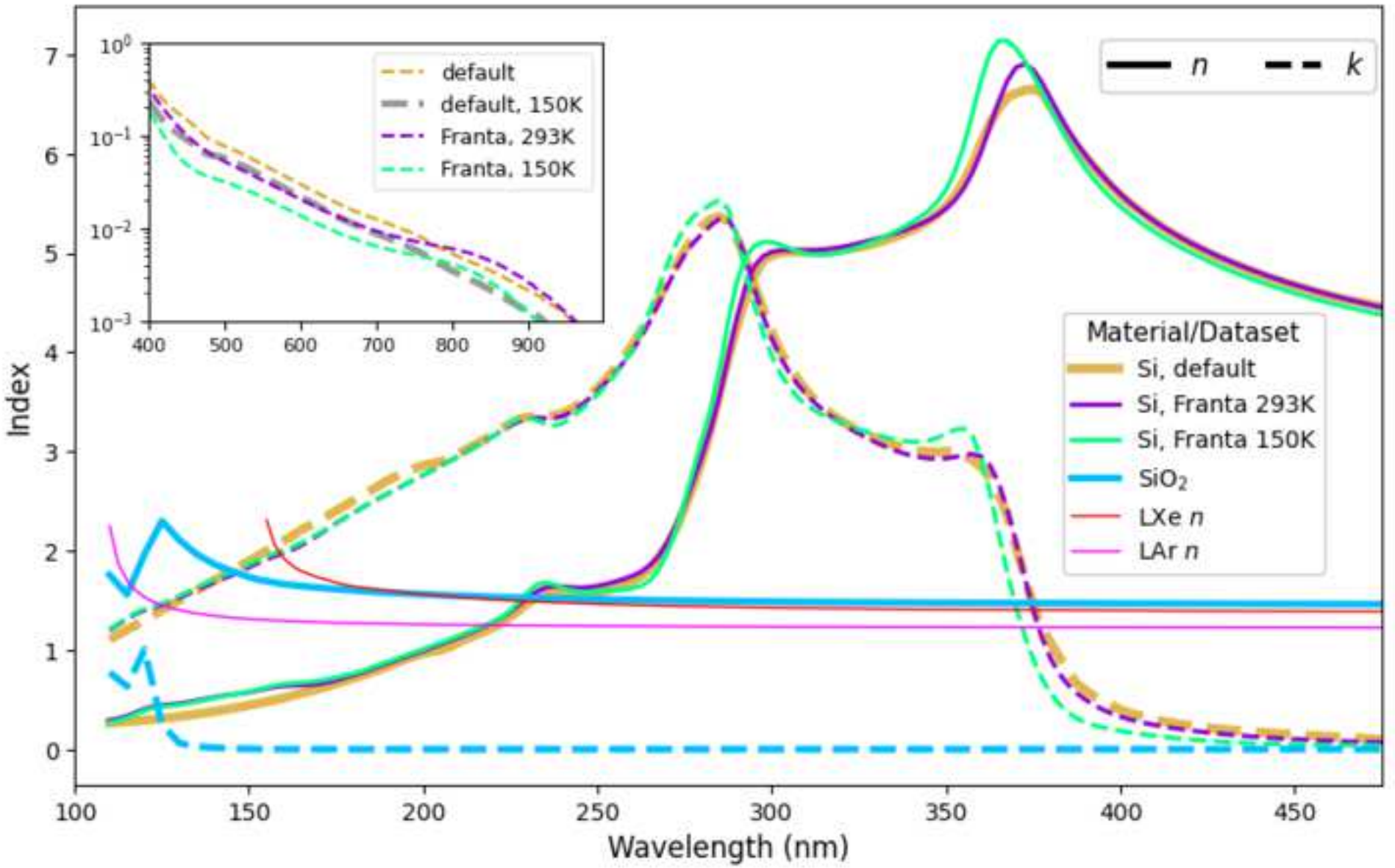




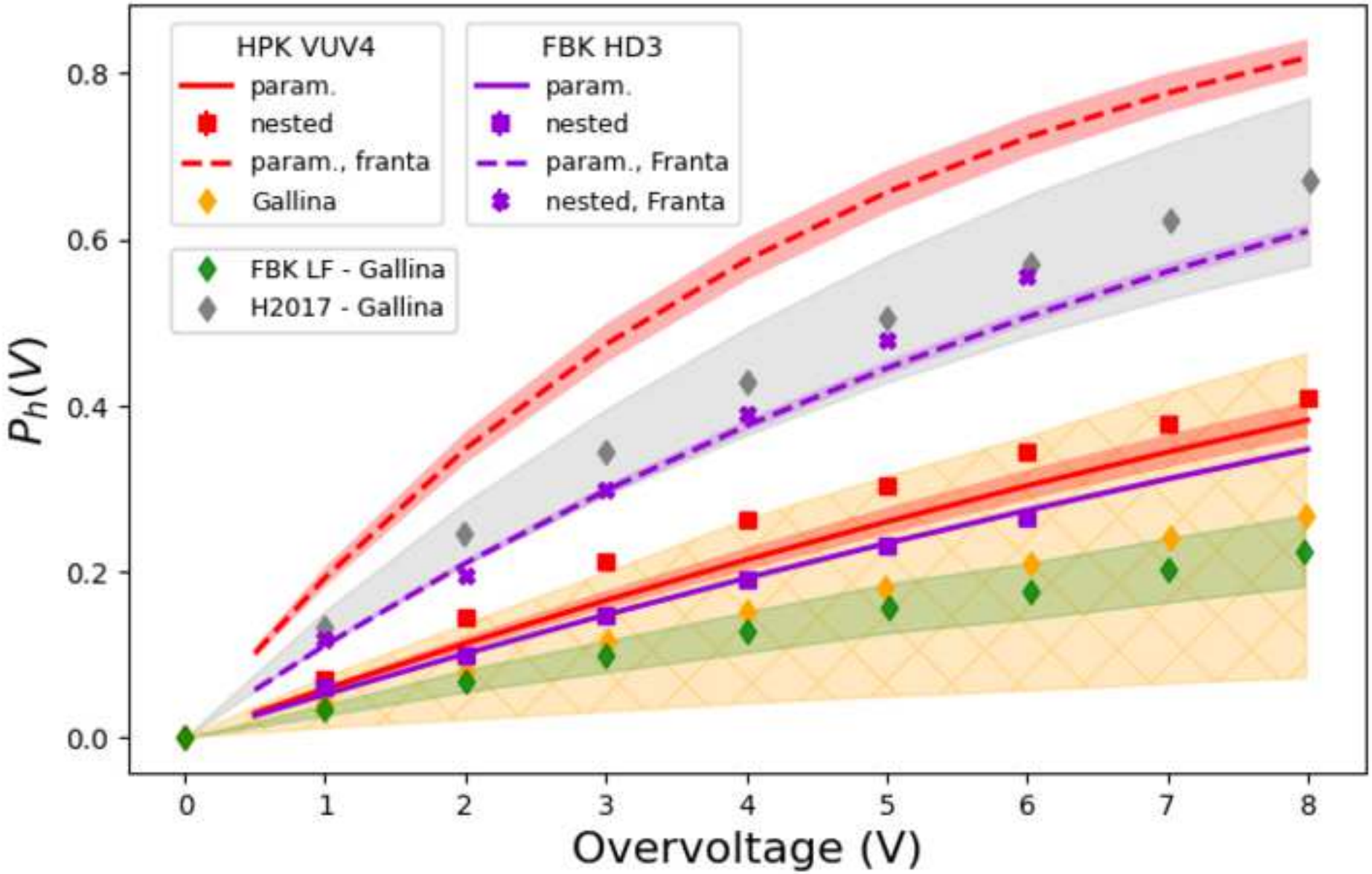
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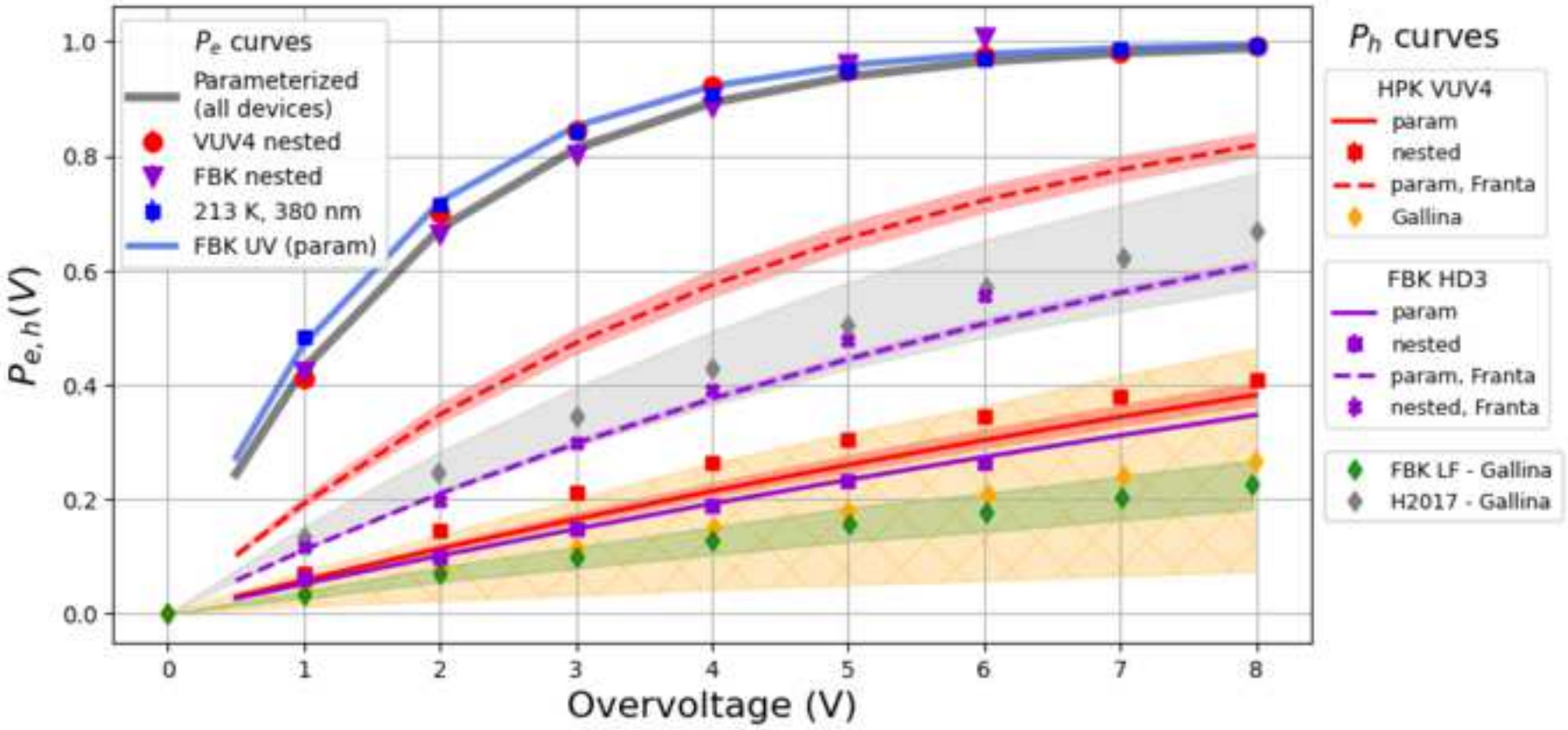
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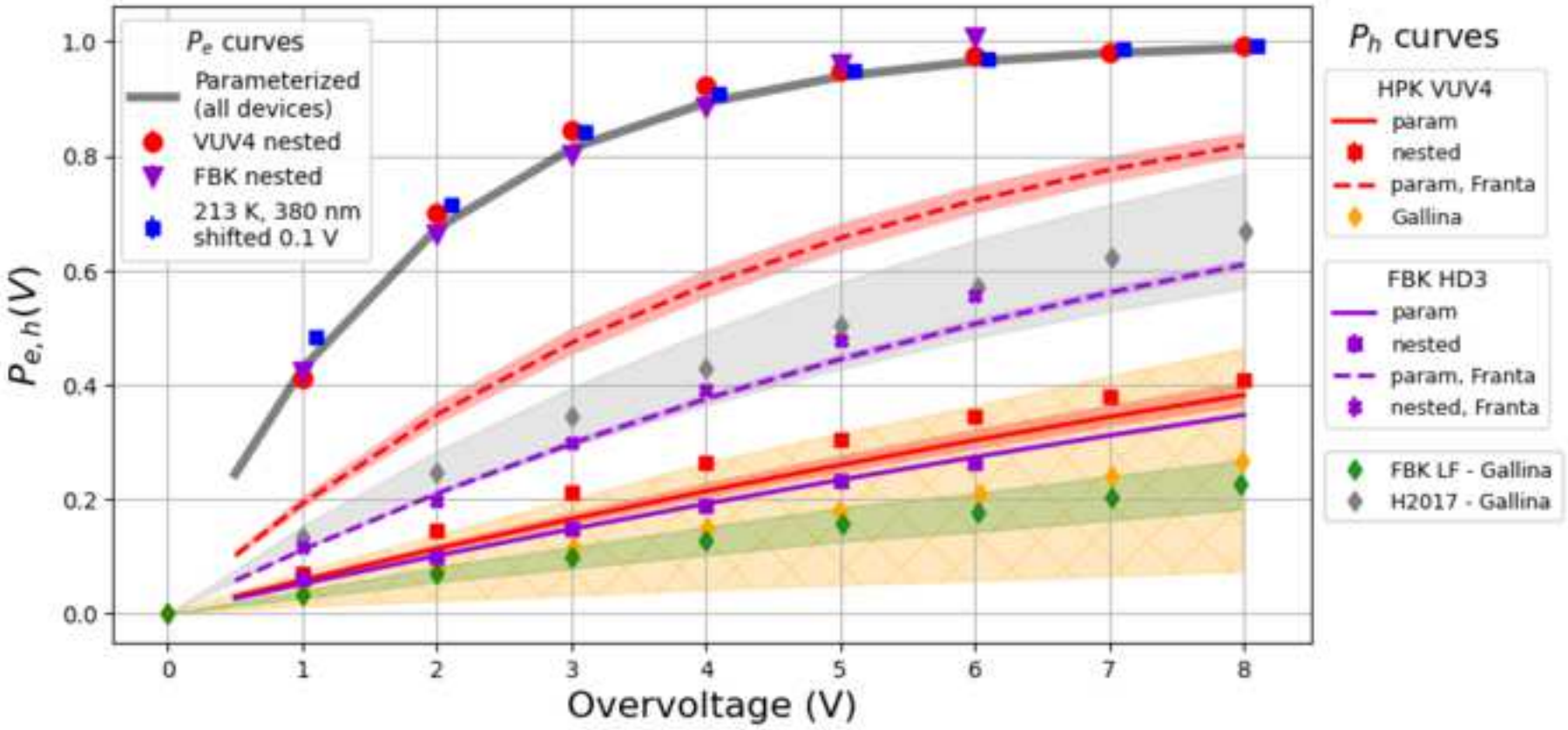
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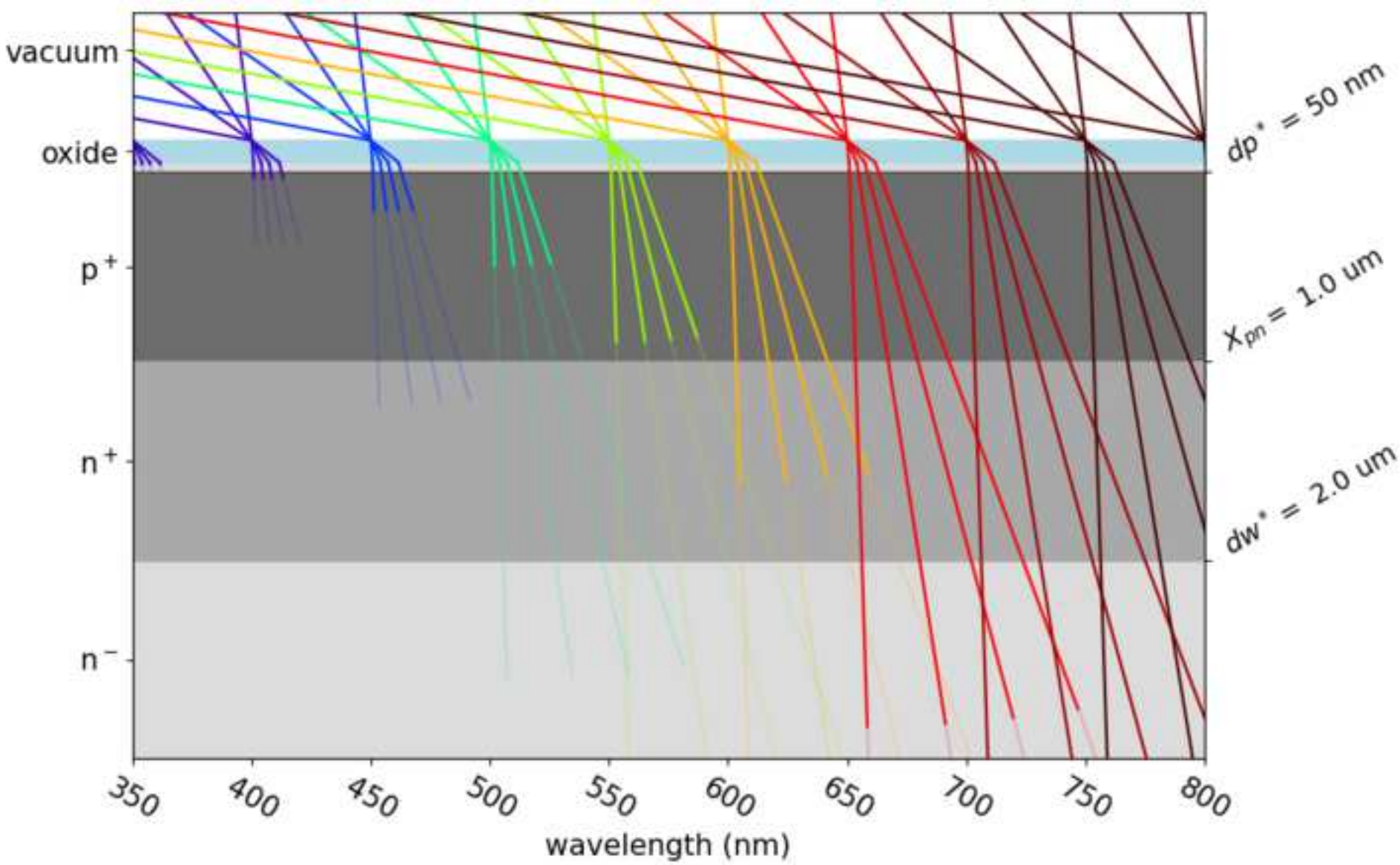
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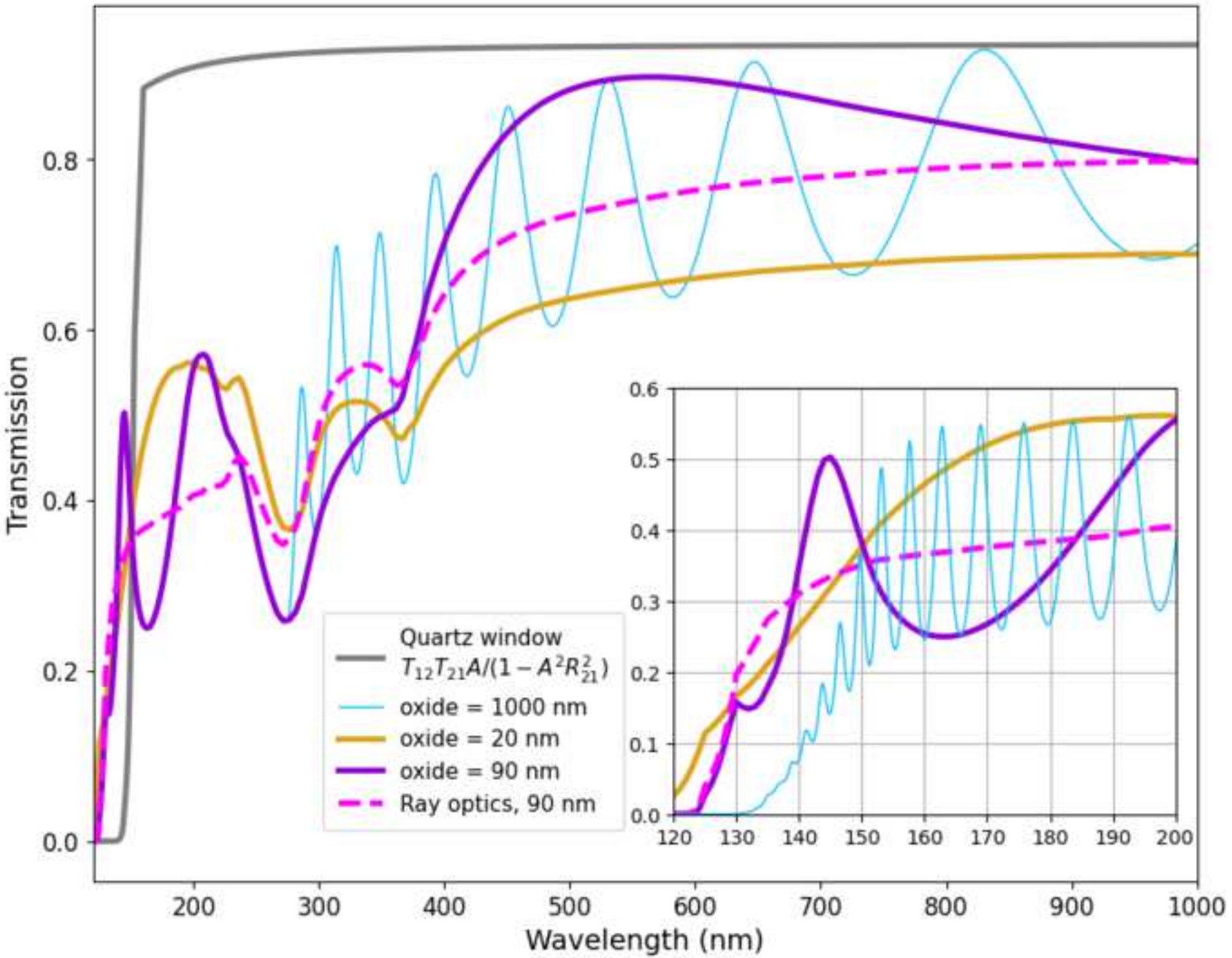
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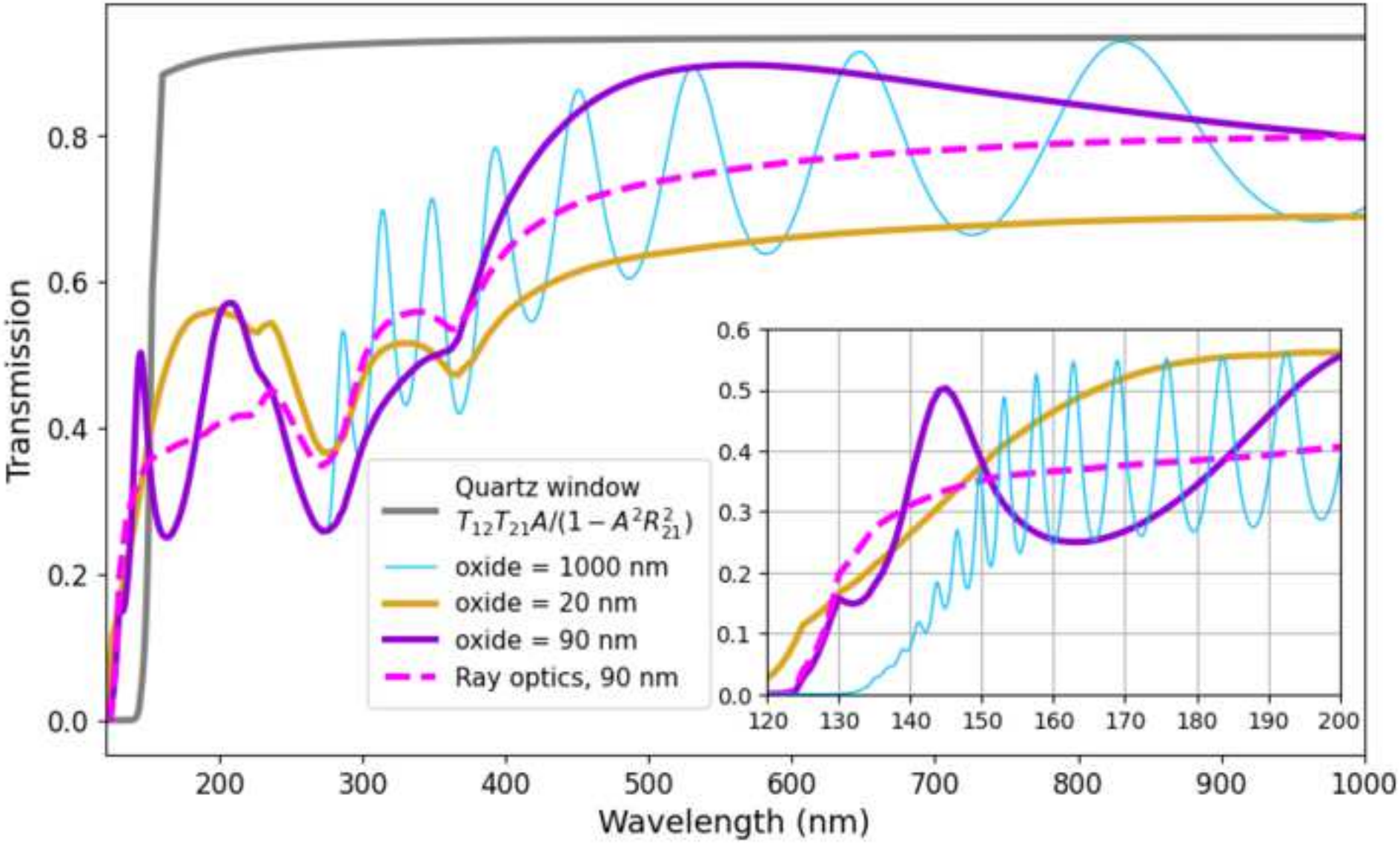
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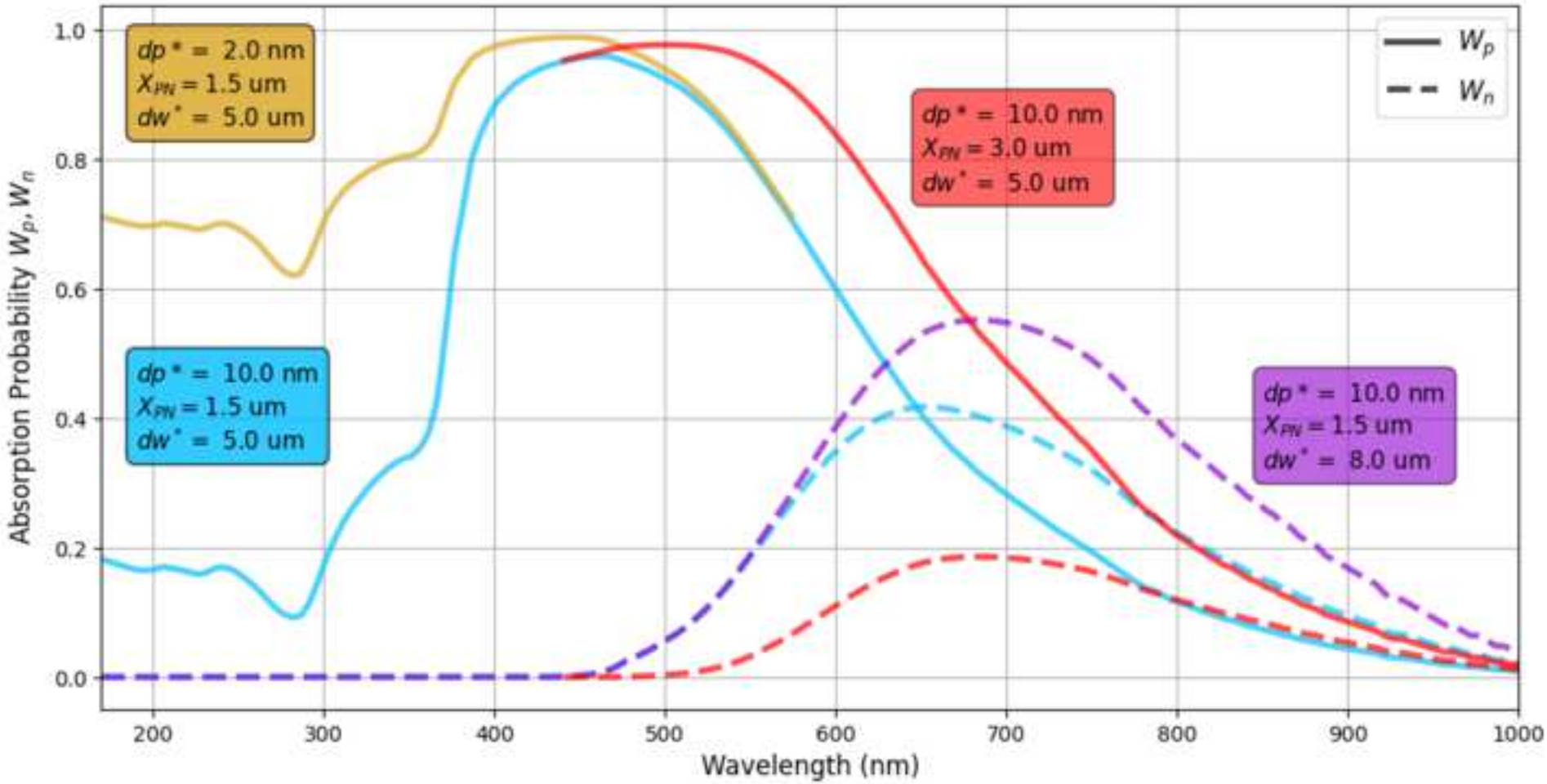
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Declaration of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.